

CZECH TECHNICAL UNIVERSITY IN PRAGUE
FACULTY OF CIVIL ENGINEERING
DEPARTMENT OF GEOMATICS

Field of Study: Geodesy and Cartography
Specialization: Geomatics



MASTER'S THESIS

Transfer learning for cloud semantic segmentation on satellite imagery

Přenosové učení pro semantickou segmentaci oblaků
na satelitních snímcích

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II. ÚDAJE K DIPLOMOVÉ PRÁCI

Název diplomové práce:

Přenosové učení pro semantickou segmentaci oblaků na satelitních snímcích

Název diplomové práce anglicky:

Transfer learning for cloud semantic segmentation on satellite imagery

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Jméno a pracoviště druhé(ho) vedoucí(ho) nebo konzultanta(ky) diplomové práce:

Datum zadání diplomové práce: **15.02.2026**

Termín odevzdání diplomové práce: **18.05.2026**

podpis vedoucí(ho) ústavu/katedry

podpis děkana(ky)

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Název diplomové práce anglicky:

Transfer learning for cloud semantic segmentation on satellite imagery

Pokyny pro vypracování:

- * Rešerše na téma přenosového učení pro satelitní snímky
- * Tvorba zdrojového a cílového datasetu
- * Učení modelu na datech Sentinel-2
- * Přenosové učení na datech VENUS
- * Porovnání s dosavadními výsledky a s odlišnými algoritmy

Seznam doporučené literatury:

- * <https://doi.org/10.1016/j.rse.2025.114694>
- * <https://doi.org/10.1016/j.rse.2024.114122>
- * <https://doi.org/10.3390/rs14205210>

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Acknowledgement

I would like to thank my family for their support throughout my studies, without which none of this would have been possible.

I am also grateful to my supervisor, Ing. Ondřej Pešek, Ph.D., for his valuable advice and flexibility throughout this thesis. Our online calls across different time zones will remain a fond memory, filled with practical tips, fun facts, and stories from around the world.

Abstract

Cloud detection is a crucial step in the processing of optical satellite imagery. Deep learning models trained for cloud detection on data from one sensor often fail when applied to a different sensor due to differences in spectral and radiometric properties. Transfer learning techniques can help mitigate this problem. This work investigates transfer learning techniques for cloud semantic segmentation, focusing on the transfer from the Sentinel-2 imagery to the VEN μ S imagery. A model based on the U-Net architecture was trained on a subset of the CloudSEN12+ dataset and subsequently adapted to a cloud detection dataset for the VEN μ S satellite. Four transfer learning configurations were compared, ranging from transductive transfer to full network fine-tuning, along with four normalization strategies. In the transductive experiment, dynamic Z-score normalization achieved the best results, enabling cross-sensor generalization without any modification of the model weights. In the fine-tuning experiments, the best results were achieved using partial adaptation with frozen encoder weights, slightly surpassing the performance of a model trained from scratch on VEN μ S data. The results demonstrate that, when the source and target domains are sufficiently similar, a model adapted through transfer learning can match or even surpass the performance of a model trained exclusively on data from the target domain. These findings suggest that transfer learning can support the deployment of cloud detection models to new sensors while decreasing the need for large, sensor-specific annotated datasets.

Key words

transfer learning, cloud detection, remote sensing, deep learning, convolutional neural network, semantic segmentation, Sentinel-2, VEN μ S

Abstrakt

Detekce oblaků je stěžejním krokem při zpracování optických satelitních snímků. Modely hlubokého učení trénované k detekci oblaků na datech z jednoho senzoru však často selhávají při aplikaci na jiný senzor kvůli rozdílům ve spektrálních a radiometrických vlastnostech. Techniky přenosového učení mohou pomoci s řešením tohoto problému. Tato práce zkoumá techniky přenosového učení pro sémantickou segmentaci oblaků se zaměřením na přenos ze snímků Sentinel-2 na snímky VEN μ S. Model založený na architektuře U-Net byl trénován na podmnožině datasetu CloudSEN12+ a následně adaptován na dataset pro detekci oblaků ze satelitu VEN μ S. Byly porovnány čtyři konfigurace přenosového učení, od transduktivního přenosu až po fine-tuning celé sítě, a zároveň čtyři normalizační strategie. V transduktivním experimentu dosáhla nejlepších výsledků dynamická Z-score normalizace, která umožnila generalizaci mezi senzory bez jakýchkoli úprav vah modelu. Z fine-tuning experimentů bylo dosaženo nejlepších výsledků za použití částečné adaptace a zmrazení vah kodéru, přičemž byl dokonce mírně překonán výkon modelu trénovaného výhradně na datech VEN μ S. Výsledky potvrzují, že pokud jsou zdrojová a cílová doména dostatečně podobné, adaptovaný model může za pomoci technik přenosového učení dosáhnout srovnatelných nebo dokonce lepších výsledků než model trénovaný výhradně na datech cílové domény. Tyto výsledky naznačují, že přenosové učení může usnadnit nasazení modelů pro detekci oblaků na nových senzorech a omezit závislost na rozsáhlých anotovaných datasetech určených pouze pro konkrétní satelit.

Klíčová slova

přenosové učení, detekce oblaků, dálkový průzkum Země, hluboké učení, konvoluční neuronová síť, sémantická segmentace, Sentinel-2, VEN μ S

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List of Abbreviations

ACCA	Automated Cloud Cover Assessment
BOA	Balanced Overall Accuracy
CNES	Centre National d'Études Spatiales
CNN	Convolutional Neural Network
ESA	European Space Agency
FN	False Negative
FP	False Positive
GMES	Global Monitoring for Environment and Security
IoU	Intersection over Union
ISA	Israeli Space Agency
MAJA	MACCS-ATCOR Joint Algorithm
mF1	Mean F1-Score
mIoU	Mean Intersection over Union
MSI	MultiSpectral Instrument
NIR	Near-Infrared
OA	Overall Accuracy
ReLU	Rectified Linear Unit
RGB	Red-Green-Blue
S2	Sentinel-2
TN	True Negative
TP	True Positive
VENμS	Vegetation and Environment monitoring on a New Micro-Satellite
VSSC	VEN μ S SuperSpectral Camera

1 Introduction

Optical satellite imagery is one of the primary sources of data for Earth observation, enabling monitoring of land cover, vegetation, urban development, and environmental change at global scale [1]. As the number of satellite missions grows and sensor resolutions increase, the volume of acquired data is expanding rapidly, making automated processing pipelines essential. Cloud detection is a particularly important prerequisite for virtually any downstream analysis of optical satellite imagery [2], and its automation is therefore critical. At the same time, the diversity of satellite sensors presents a significant challenge for deep learning methods, which have become widely adopted in many remote sensing tasks [3, 4]. These methods often require large amounts of labeled training data, which are expensive and time-consuming to produce [5], and models trained on one sensor often do not generalize well to another [4]. Transferring cloud detection knowledge from a well-established sensor to a new one could reduce or even eliminate the need for labeled target domain data, significantly lowering the effort required to develop cloud detection methods for new satellite missions.

Transfer learning has been explored for cross-sensor cloud detection in a limited number of works [4, 6]. Existing approaches have either focused on developing sensor-agnostic models through data augmentation and normalization, or applied transfer learning without systematically exploring the range of available adaptation strategies. The combined effect of fine-tuning strategies and normalization approaches in cross-sensor transfer scenarios remains underexplored.

This thesis addresses this gap by investigating cross-sensor transfer learning for cloud semantic segmentation on satellite imagery, with a focus on systematically comparing fine-tuning strategies and normalization approaches for the transfer from Sentinel-2 imagery to VEN μ S (Vegetation and Environment monitoring on a New Micro-Satellite) imagery. Sentinel-2 is a well-established Earth observation mission with large publicly available labeled datasets, while VEN μ S was a smaller research mission with a similar but not identical spectral configuration, making it a representative target. The goals of this thesis are to review and synthesize the theoretical foundations of cross-sensor transfer learning for cloud detection, to prepare and preprocess source and target datasets, to train and evaluate a deep learning model under multiple transfer learning and normalization configurations, and to compare the transfer learning results against a from-scratch deep learning baseline and a classical cloud detection algorithm. The study is guided by three research questions:

1. Can a U-Net model trained on Sentinel-2 data generalize to VEN μ S imagery using only spectral band alignment as a base domain adaptation, and how does additional input-level radiometric normalization affect this generalization?
2. How much of the pretrained network needs to be adapted to achieve effective cross-sensor transfer, and does fine-tuning the full network outperform partial adaptation strategies?
3. Can transfer learning from Sentinel-2 achieve performance comparable to a model trained from scratch on VEN μ S data, and if so, what degree of adaptation is required to achieve this?

Beyond the methodological contribution, the work has direct practical relevance. Although the VEN μ S mission has concluded, its archive of imagery remains a valuable resource for retrospective research, and generating improved cloud masks for all historically acquired scenes would significantly facilitate their use in future studies. The same approach could further be applied to other satellite missions for which annotated cloud detection data are scarce or of limited quality.

The remainder of this thesis is structured as follows. Chapter 2 provides the theoretical background necessary for understanding cloud detection, relevant deep learning and transfer learning techniques, and the evaluation metrics used in this work. In Chapter 3, the datasets used, preprocessing pipeline, experimental setup, and implementation are described. Chapter 4 presents the experimental results and addresses the three research questions. In Chapter 5, the findings are interpreted and compared with related work, and limitations and directions for future research are identified. Finally, Chapter 6 summarizes the main conclusions and contributions of the thesis.

2 Theoretical Background

This chapter provides the theoretical background necessary for understanding the methods and experimental design used in this thesis. Section 2.1 introduces cloud detection, its importance in optical satellite imagery analysis, and the main challenges associated with it. Section 2.2 reviews the most common cloud detection methods, with a focus on deep learning approaches and the U-Net architecture. Section 2.4 describes the principles of transfer learning, including the formal definitions of transductive and inductive settings and common fine-tuning strategies. Finally, Section 2.5 introduces the evaluation metrics and Section 2.6 the loss functions used throughout the experimental part of the thesis.

2.1 Cloud Detection

Since approximately two-thirds of the Earth’s surface is covered by clouds at any given time [7], cloud detection is an essential preprocessing step in the analysis of optical satellite imagery [2]. It is typically formulated as a semantic segmentation problem, where each pixel in a satellite image is labeled as either cloud or cloud-free, or, in some approaches, cloud, cloud shadow, or clear [5]. The presence of clouds prevents optical satellite imaging systems from obtaining useful observations by blocking the view of the Earth’s surface. Accurate identification of cloudy pixels is, therefore, crucial in many types of remote sensing applications, including land cover monitoring, disaster response, and military surveillance [5, 1].

There are several challenges that complicate precise cloud detection. First, clouds can exhibit spectral reflectance similar to bright surfaces such as snow, desert areas, or some urban regions, which may lead to misclassifications. In addition, there is a high variability in cloud types. According to [1], clouds can be categorized based on their height above ground level or based on optical density. On the basis of height, clouds are typically sorted into low, medium, and high categories. Low clouds are found below approximately 2 km and are in most cases optically thick, often completely blocking the view of Earth’s surface and producing strong shadows. Medium clouds are found between 2 km and 6 km and are slightly less dense than low clouds, partially limiting surface visibility. High clouds occur above 6 km, are mostly optically thin, and the surface under them remains visible, although often blurry [8]. While cloud altitude provides a helpful way to categorize clouds, their optical density also plays an important role in order to classify clouds automatically [5]. Thick clouds have opacity above 50%, obscure the surface and cast dark shadows. These shadows are an additional challenge in automated cloud detection, due

to their appearance being influenced by cloud shape, height, thickness, sun angle, and surface properties [1]. Cloud shadows can be mistaken for water bodies, shaded areas or other dark regions, making their accurate identification also essential [9]. Thin clouds, on the other hand, have opacity under 50% and allow partial visibility of the surface [1]. Thin clouds can also cast shadows, although these are usually fainter and even more difficult to detect [2]. A comparison of thick and thin clouds is shown in Figure 2.1.

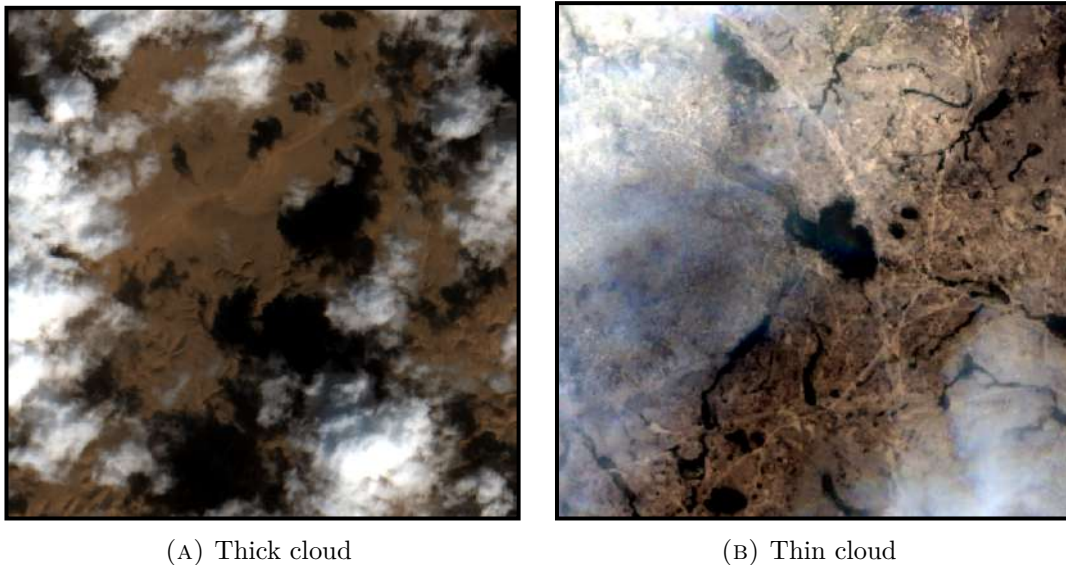


FIGURE 2.1: Comparison of thick and thin clouds in optical satellite imagery.

2.2 Methods for Cloud Detection

Given the complexity and significance of the problem and the challenges described above, a wide range of cloud detection methods and algorithms have been proposed over the last decades. These methods continue to evolve in order to improve detection accuracy and further automate the process. Various classifications of these methods have been proposed in the literature, reflecting differences in input data, algorithmic approach, and level of automation.

Cloud detection methods in the literature can generally be classified along two main criteria. The first criterion considers the algorithmic approach and level of automation, while the second criterion focuses on the type of input data and the features exploited by the algorithms. These two perspectives often overlap, and the choice of classification depends on the focus of the study and the available data.

Focusing on the algorithmic criterion, Singh et al. [1] divide methods into manual, automated, and active learning approaches. Even though manual methods are generally the most accurate, they require the most effort and are primarily used today for ground truth dataset creation. Automated methods

include both rule-based and machine learning approaches, which operate without human intervention. Active learning represents a specialized approach, often based on machine learning, that uses human feedback to train and refine the algorithm's predictions. A more common distinction, used for example by Wang et al. [5], categorizes cloud detection methods into physical-rule or threshold-based, temporal-change-based, and machine learning-based algorithms. Li et al. [2] additionally include variational-model-based methods as a separate category of cloud detection algorithms.

Considering the type of input data, cloud detection methods can be classified according to the features extracted and the temporal coverage of the imagery. Li et al. [2] further classify cloud and cloud-shadow detection algorithms based on the type of features used. Methods can rely on spectral features, spectral-spatial features, spectral-temporal features, spectral-spatial-temporal features, or multi-source features, which can be extracted from mono-temporal, multi-temporal, or multi-source data. Singh et al. [1] further classify cloud detection methods based on the number of spectral bands used. Single-band approaches rely on one dominant band sensitive to clouds, while multi-band approaches combine several bands (e.g., red-green-blue (RGB), near-infrared (NIR)) to improve detection accuracy.

The classification based on the algorithmic approach and level of automation proposed by Wang et al. [5] is adopted, as it covers all three algorithm-based categories relevant to this work.

2.2.1 Threshold-Based Methods

Threshold-based algorithms represent one of the earliest and most widely used approaches for cloud detection [5]. These algorithms perform cloud detection by applying predefined thresholds to one or more spectral bands in order to classify individual pixels [2, 1]. These thresholds are typically determined based on characteristic cloud properties, such as high reflectance in visible bands or specific spectral behavior in selected wavelengths [5]. Due to their simplicity and low computational requirements, these methods have been widely used in many operational systems [10].

However, a major limitation of many threshold-based approaches is their dependence on empirically defined parameters. Thresholds are defined for specific sensors, regions, and atmospheric conditions and for this reason, the algorithms are often applicable only to specific sensor or sensors similar to the original one [2]. Several algorithms have been developed for different satellite missions [5, 10], including the Automated Cloud Cover Assessment (ACCA) for Landsat imagery [11, 12], as well as Sen2Cor for Sentinel-2 imagery [13] and Cloud-Mask for MODIS imagery [14]. To improve applicability, some methods have been extended to support multiple satellite platforms. For example, FMask was originally developed for Landsat [15] and later adapted for Sentinel-2 [16].

Some algorithms, such as MACCS-ATCOR Joint Algorithm (MAJA), use threshold-based techniques in combination with temporal information to improve detection. The details of such multi-temporal approaches are discussed in section 2.2.2.

2.2.2 Temporal-Change-Based Methods

Temporal-change-based algorithms achieve cloud detection by identifying changes in reflectance caused by clouds or cloud shadows. By analyzing multiple views of the same location over time, these methods are particularly effective at distinguishing clouds from bright surfaces such as urban areas [2]. An example of a temporal-change-based algorithm is MAJA, which is the default cloud-masking algorithm for VEN μ S imagery, but it is also possible to use MAJA on Formosat-2, Landsat, and Sentinel-2 data [17]. MAJA combines spectral thresholding with multi-temporal analysis, using image time series to improve cloud and cloud-shadow detection [10, 17].

Overall, temporal-change-based approaches offer enhanced accuracy compared to threshold-based methods, but they require multi-temporal data and additional processing [2, 10].

2.2.3 Machine Learning-Based Methods

Machine learning-based methods have been widely applied to cloud detection as an alternative to threshold-based approaches [1, 5]. These methods learn from the input data and iteratively optimize their performance. They can be divided into supervised and unsupervised approaches. Supervised methods learn the relationship between input features and cloud labels from annotated training data, allowing them to adapt to different cloud types, surface types and lighting conditions. On the other hand, unsupervised methods try to identify patterns or clusters in the data without labeled samples, for example by grouping pixels with similar spectral properties.

Even though traditional supervised machine learning methods, such as decision trees, random forests and support vector machines can learn from the input data, they are limited in their ability to capture complex spatial patterns. This is because they typically operate on spectral features extracted from individual pixels. It is possible to include additional spatial features to improve detection accuracy, but this requires explicit feature engineering, which is particularly challenging for clouds due to their highly variable shapes and appearances, and thus leads to increased workload [1, 2].

This limitation has led to the rapid development and adoption of deep learning approaches [5]. In particular, convolutional neural networks (CNNs) have proven to be efficient at capturing both spectral and spatial features from raw images [10, 3]. More recently, hybrid approaches combining CNNs with transformer-based architectures have also been proposed to further improve performance [18, 19]. However, due to higher complexity of hybrid approaches, this work focuses on the usage of CNNs, possibly serving as a springboard for a

hybrid-focused extension. A description of the CNN principle and an overview of CNN-based methods for cloud detection are provided in section 2.3.

2.3 Convolutional Neural Networks

Convolutional neural networks (CNNs) are a class of deep learning networks specifically designed for processing data with grid-like topology, such as images [20, 21]. CNNs exploit the spatial structure of the input data, allowing neurons in each layer to connect only to local regions rather than to all neurons in the previous layer. This local connectivity reduces the number of parameters and enables the network to learn spatially localized patterns efficiently [21]. CNNs can automatically learn hierarchical features from input images primarily through convolutional layers and pooling layers [20].

Convolutional layers apply a fixed-size kernel that slides over the input, computing a weighted sum between the kernel's weights and the local neighborhood of each pixel [20]. The principle of convolution is illustrated in Figure 2.2. Since convolution typically reduces the dimensions of the output relative to the input, padding is often applied around the input data to maintain the spatial size of the feature maps [20]. In remote sensing, convolution is typically performed over multiple spectral bands. For multispectral data, the kernels are three-dimensional tensors, where the depth of the kernel is equal to the number of spectral bands [3].

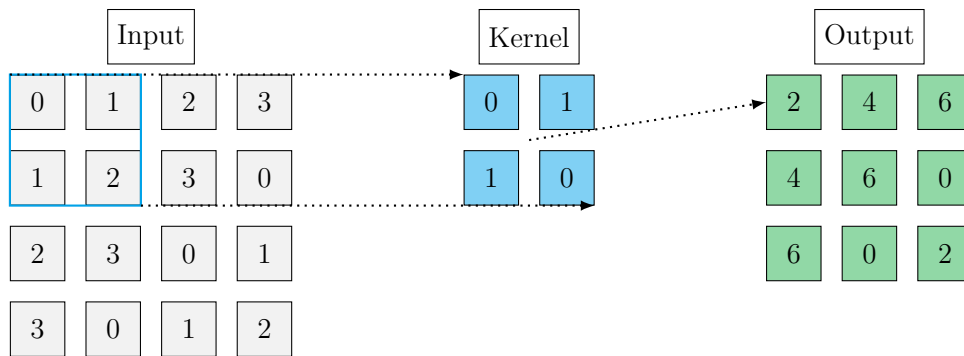
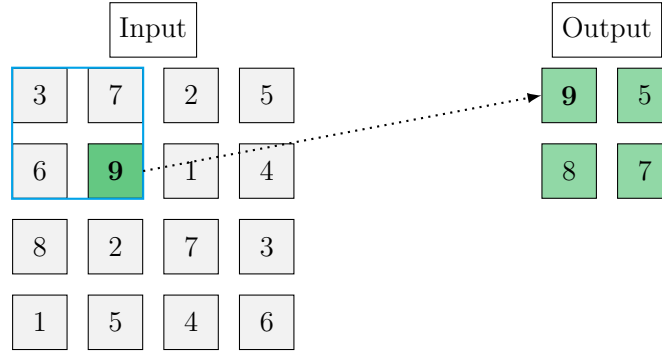


FIGURE 2.2: Principle of 2D convolution.

The linear activations produced by the convolution are then passed through a nonlinear activation function, typically a rectified linear unit (ReLU) [20]. The ReLU function sets all negative values to zero while leaving non-negative values unchanged.

Pooling layers reduce the spatial resolution of feature maps while preserving the most important information [22]. The most commonly used type is max pooling, where the maximum value is selected from a defined window, representing the most significant feature in that region (see Figure 2.3).

FIGURE 2.3: Principle of 2×2 max pooling with stride 2.

In addition to architectural components, the training of CNN models is strongly influenced by the choice of loss function. In semantic segmentation, losses such as Dice loss are often used instead of standard cross-entropy, as they are better suited for handling class imbalance [10].

In recent years, CNNs have proven highly effective for image detection and segmentation in remote sensing. Several studies have successfully applied them to cloud detection in satellite imagery [2, 5, 10, 4, 23, 24]. Typically, researchers have either developed architectures specifically for cloud detection or adapted popular models from other domains. The following section lists the most commonly used CNN architectures for cloud detection and describes the one relevant for this study.

2.3.1 CNNs for Cloud Detection

According to multiple studies [2, 5, 10, 23, 25], commonly used architectures for cloud detection include segmentation models such as U-Net [26], SegNet [27], and DeepLabV3 [28]. In addition, backbone architectures such as VGG [29] and ResNet [30] are often used as encoders, where they serve as feature extractors. Among the various approaches, U-Net [26] has become particularly popular, with its numerous modifications consistently achieving state-of-the-art performance in remote sensing and cloud detection tasks [5]. It has also been shown to perform best for cloud detection on VEN μ S imagery [10], which is the primary reason it was selected for this work.

U-Net

U-Net, introduced by Ronneberger et al. [26] is a fully convolutional neural network originally designed for biomedical image segmentation. The main idea of U-Net is based on an encoder-decoder structure with a symmetric U-shaped design (see Figure 2.4). Due to its effectiveness and simplicity, this architecture has become one of the most widely adopted approaches for semantic segmentation in the field of computer vision.

The encoder, also called the contracting path in the original paper, uses a combination of convolutional layers, ReLU activation functions, and pooling

layers, as described in Section 2.3, to extract relevant features from the input image and reduce its spatial resolution. The decoder, also called expansive path, gradually restores the spatial resolution using upsampling operations and convolutional layers. A key part of U-Net is the use of skip connections. These connections transfer feature maps directly from the encoder to the corresponding decoder layers. This helps to maintain spatial detail, which is particularly important in segmentation tasks where accurate boundary estimation is required, such as cloud detection.

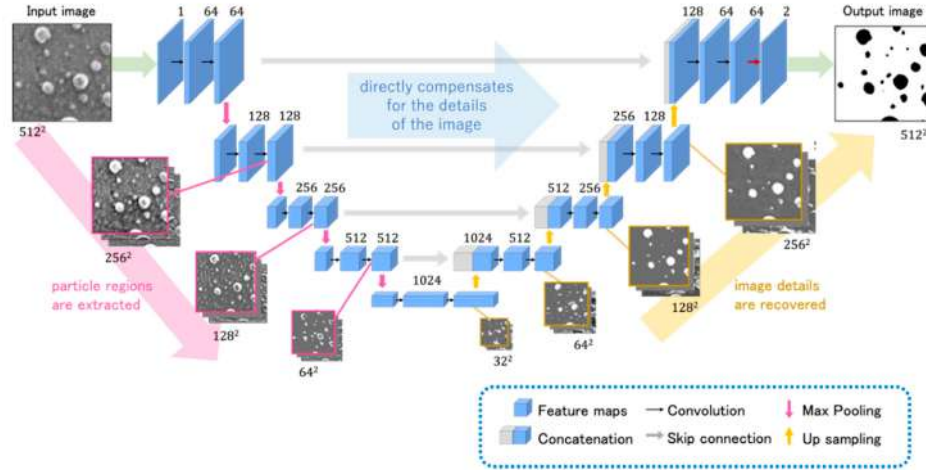


FIGURE 2.4: U-Net architecture.

Source: [31]

2.3.2 Challenges in CNN-Based Cloud Detection

As described in Section 2.1, one of the main challenges in cloud detection is the complexity of clouds themselves. Clouds can vary significantly in shape, thickness, and transparency [1]. Handling these variations therefore requires a robust model trained on a diverse and representative dataset [5].

Another important issue is class imbalance, which is common in cloud detection datasets [32]. In most cases, the number of cloud-free pixels is significantly larger than the number of cloud pixels. This can lead to biased training, where the model tends to favor the dominant class and may fail to correctly identify less frequent cloud regions. To address this issue, Dice loss function [33] is often used, as it helps to reduce the impact of class imbalance during training [10], as described in Section 2.6.

CNN models are also sensitive to the quality and geographic distribution of training data [34]. Models trained on limited or region-specific data may not generalize well to unseen conditions or different geographic areas. This motivates the distinction between globally trained models, and locally trained models, which are optimized for a specific geographic region. This issue is

closely related to the problem of domain shift. In remote sensing, data acquired from different satellites often vary in spatial resolution, spectral characteristics, and radiometric properties [2, 4]. As a result, models trained on data from one satellite may experience a drop in performance or even fail entirely when applied to data from a different sensor [4].

To address these challenges, two main approaches are commonly used in remote sensing applications. The first approach focuses on developing more robust models by training on diverse and representative datasets, often combined with data augmentation techniques to improve generalization across different conditions and sensors. In this context, Wright et al. proposed OmniCloud-Mask [4], a sensor-agnostic cloud and cloud-shadow detection model based on the U-Net architecture. Although the model was trained only on Sentinel-2 data, the authors used multiple augmentation techniques, including flipping, rotation, cropping, and, most importantly, mixed-resolution training, together with dynamic Z-score normalization to reduce spectral differences between sensors. This enabled a model trained on Sentinel-2 data to generalize successfully to Landsat 8 and PlanetScope imagery.

The second approach is based on transfer learning, which aims to adapt a model trained on a source domain to a different but related target domain, and is particularly effective when source and target domains differ in their data distributions [20]. An example was presented by Mateo-García et al. [6], who investigated transferring cloud detection model based on U-Net architecture between Landsat 8 and Proba-V sensors. Their experiments addressed three key questions related to cross-sensor transfer: adaptation from Landsat 8 to Proba-V, transfer from lower-resolution Proba-V to higher-resolution Landsat 8, and the effect of combining source and target domain data on model accuracy. The results confirmed that transfer learning can be effective in both directions and that incorporating limited target-domain data further improves performance, reducing the need for large manually labeled datasets. In the following chapter, the fundamental principles and common techniques of transfer learning are described in more detail.

2.4 Transfer Learning

As introduced in the previous chapter, transfer learning leverages knowledge acquired from a source domain to improve learning in a different but related target domain [20].

The most widely adopted formal definition was introduced by Pan and Yang [35]. Given a source domain $\mathcal{D}_S$ with learning task $\mathcal{T}_S$ and a target domain $\mathcal{D}_T$ with learning task $\mathcal{T}_T$, transfer learning aims to help improve the learning of the target predictive function $f_T(\cdot)$ in $\mathcal{D}_T$ using the knowledge in $\mathcal{D}_S$ and $\mathcal{T}_S$, where $\mathcal{D}_S \neq \mathcal{D}_T$, or $\mathcal{T}_S \neq \mathcal{T}_T$.

A domain is defined as a pair $\mathcal{D} = \{\mathcal{X}, P(X)\}$, consisting of a feature space $\mathcal{X}$ and a marginal probability distribution $P(X)$. The condition $\mathcal{D}_S \neq \mathcal{D}_T$ implies that the feature spaces differ ($\mathcal{X}_S \neq \mathcal{X}_T$), the marginal distributions differ

($P_S(X) \neq P_T(X)$), or both. In the context of remote sensing and cloud detection, this corresponds to the situation where two sensors differ in their spectral bands or spatial resolution ($\mathcal{X}_S \neq \mathcal{X}_T$), where the same spectral features follow different radiometric distributions across sensors ($P_S(X) \neq P_T(X)$), or both, as may occur when comparing sensors that differ in both their band configurations and calibration.

A task is defined as a pair $\mathcal{T} = \{\mathcal{Y}, P(Y|X)\}$, consisting of a label space $\mathcal{Y}$ and a conditional distribution $P(Y|X)$. The condition $\mathcal{T}_S \neq \mathcal{T}_T$ implies that either the label spaces differ ($\mathcal{Y}_S \neq \mathcal{Y}_T$) or the conditional distributions differ ($P(Y_S|X_S) \neq P(Y_T|X_T)$), or both. In cloud detection, the former arises when the set of predicted classes differs between sensors, for instance, when a cloud-shadow class is present in the source domain, but absent in the target domain. The latter occurs when the relationship between spectral observations and land cover types changes due to geographic or environmental factors. For example, a specific high-reflectance signature may represent a cloud in the source domain but correspond to snow or ice in a high-altitude target domain.

Pan and Yang [35] categorize transfer learning into three settings based on the relationship between source and target domains and tasks: transductive transfer learning, inductive transfer learning, and unsupervised transfer learning. These settings are summarized in Figure 2.5.

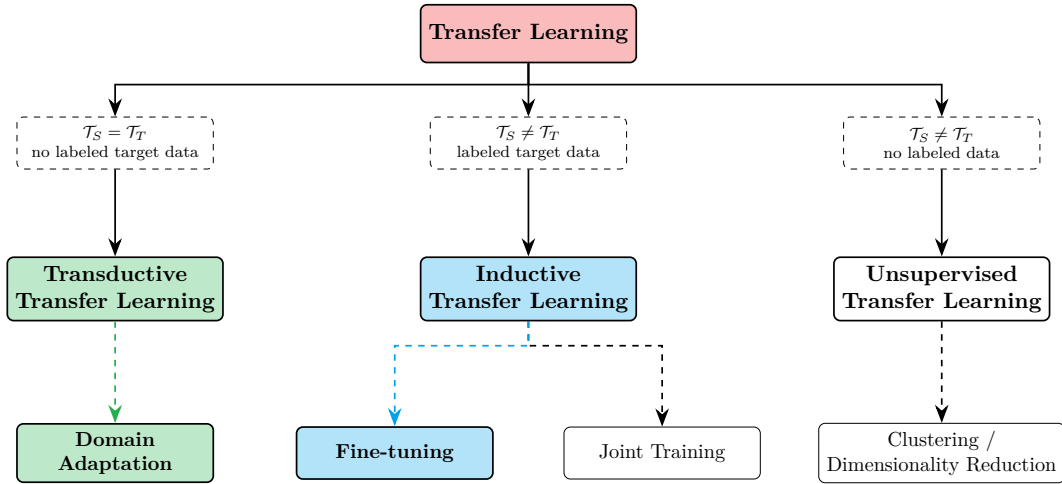


FIGURE 2.5: Overview of transfer learning settings and selected related techniques.

In transductive transfer learning, the source and target tasks are the same ($\mathcal{T}_S = \mathcal{T}_T$), while the domains differ ($\mathcal{D}_S \neq \mathcal{D}_T$). No labeled data in the target domain are required. A representative technique is domain adaptation, which aims to reduce the discrepancy between source and target domains so that a model trained on the source can be applied to the target without any labeled target data [36]. In inductive transfer learning, the target task differs from the source task ($\mathcal{T}_S \neq \mathcal{T}_T$), and labeled data in the target domain are available to help with adaptation [35]. When labeled source data are also available,

this setting is commonly realized either by fine-tuning, which uses the source-domain model as initialization for subsequent training on target data, or by joint training, which combines source and target data into a single training set [6]. In unsupervised transfer learning, no labeled data are available in either domain and the focus is on unsupervised objectives such as clustering or dimensionality reduction [35].

In remote sensing, cross-sensor adaptation without any target labels corresponds to the transductive setting, while the availability of labeled target data shifts the problem to the inductive setting. Mateo-García et al. [6] investigated both settings in the context of cloud detection, transferring a U-Net-based model between Landsat-8 and Proba-V. In the transductive experiments, models were trained on one sensor and evaluated on the other without any target-domain labels. In the inductive experiments, they compared both joint training and fine-tuning.

Fine-tuning can be applied either to the whole network or only to selected parts. A common fine-tuning strategy is to freeze selected layers of the pre-trained network, meaning their weights are held fixed during training and receive no gradient updates. Yosinski et al. [37] experimentally showed that shallower layers of deep neural networks usually learn more general features that can be transferred across tasks and domains, while deeper layers are more specialized to the original task. Therefore, it is often suggested to freeze earlier layers and adapt only the later layers. This is especially useful when the target dataset is small compared to the number of trainable parameters, as updating fewer weights can help reduce overfitting.

In encoder-decoder architectures such as U-Net, this often corresponds to using pretrained weights for the encoder while fine-tuning the decoder and the final output layers. If sufficient target-domain data are available, the whole network can also be fine-tuned, typically with a lower learning rate.

When the target dataset uses a different set of classes than the source dataset, the final classification layer usually needs to be replaced or reinitialized so that it matches the target labels, while the remaining network parameters can still be transferred.

In this thesis, transfer learning is studied in a cross-sensor remote sensing setting, where models trained on one satellite dataset are adapted to a different target dataset acquired from another sensor. Differences in spatial resolution, spectral configuration, and radiometric properties introduce a domain shift between the datasets.

2.5 Evaluation Metrics

Evaluation metrics are used to quantify the performance of classification and segmentation models and to allow comparison between different methods. Visual inspection is subjective and not suitable for large-scale evaluation, so quantitative metrics are used to measure the performance of the method used.

Evaluation is typically based on the confusion matrix [38], which provides an overview of the relationship between predicted and reference (ground truth) labels at pixel level. The confusion matrix aggregates model predictions over all evaluated pixels and captures how individual classes are confused with each other. Since the binary case is merely a simplification of the multiclass formulation, the multiclass case is explained here. Each pixel is assigned to one of the classes, such as cloud, clear, or cloud shadow. An example of the confusion matrix is shown in Figure 2.6.

Reference label	Cloud	TP	FN	FN
	Clear	FP	TN	TN
	Shadow	FP	TN	TN
		Cloud	Clear	Shadow
		Predicted label		

Highlighted: Cloud class (one-vs-rest)

FIGURE 2.6: Multi-class confusion matrix with one-vs-rest interpretation for the Cloud class.

In the multiclass setting, metrics are commonly computed for each class using a one-vs-rest interpretation. For a selected class, all remaining classes are treated as the negative class, which allows the computation of true positives (TP), false positives (FP), false negatives (FN), and true negatives (TN). The highlighted region in Figure 2.6 illustrates this one-vs-rest formulation for the cloud class, where all remaining classes (clear and cloud shadow) are treated as the negative class.

In this work, confusion matrices are normalized by row, meaning each value represents the proportion of pixels of the true class predicted as each class. Diagonal values therefore correspond to per-class recall (explained below), and the total number of pixels per class is reported alongside the matrix for reference.

From the confusion matrix, a range of commonly used evaluation metrics can be derived to quantify model performance [38], and the most commonly used of these metrics are described below.

Precision, also known as user’s accuracy [32], expresses how reliable the predictions of a given class are. It is defined as the ratio between correctly predicted samples of the class and all samples predicted as that class:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2.1)$$

Recall, also referred to as sensitivity or producer's accuracy in remote sensing literature [32], measures the ability of the model to correctly identify all pixels belonging to the given class:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2.2)$$

Overall Accuracy (OA) expresses the proportion of correctly classified pixels among all evaluated pixels. In the multiclass case, it is computed as the sum of correctly classified samples on the main diagonal of the confusion matrix divided by the total number of samples:

$$\text{OA} = \frac{\sum_i CM_{ii}}{\sum_i \sum_j CM_{ij}}, \quad (2.3)$$

where CM_{ij} denotes the element of the confusion matrix at row i and column j .

Although OA is intuitive and easy to interpret, it can be misleading in the presence of class imbalance, where high values may be achieved by favoring dominant classes such as clear pixels.

Balanced Overall Accuracy (BOA) addresses this limitation by averaging the recall across all classes, giving equal weight to each class regardless of its frequency:

$$\text{BOA} = \frac{1}{C} \sum_{c=1}^C \text{Recall}_c, \quad (2.4)$$

where C denotes the number of classes.

Unlike OA, BOA is not biased towards dominant classes and provides a more reliable measure of performance for imbalanced datasets.

F1-score combines precision and recall into a single metric using their harmonic mean, giving equal weight to false positives and false negatives. It is therefore particularly useful when working with imbalanced datasets:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (2.5)$$

Mean F1-score (mF1) is the arithmetic mean of per-class F1-scores:

$$\text{mF}_1 = \frac{1}{C} \sum_{c=1}^C F_{1,c}, \quad (2.6)$$

where C denotes the number of classes.

By averaging over all classes, the mF1 gives equal importance to minority classes, such as cloud shadow.

Intersection over Union (IoU), also known as the Jaccard index, measures the overlap between predicted and reference regions for a given class:

$$\text{IoU} = \frac{TP}{TP + FP + FN} \quad (2.7)$$

Unlike the F1-score, which represents a harmonic mean of precision and recall, IoU directly measures the overlap between predicted and reference regions.

For multiclass segmentation, the IoU is often averaged across all classes and is called the **mean Intersection over Union (mIoU)**:

$$\text{mIoU} = \frac{1}{C} \sum_{c=1}^C \text{IoU}_c, \quad (2.8)$$

where C denotes the number of classes.

By averaging over all classes, mIoU gives equal importance to minority classes, such as cloud shadow, just as mF1.

In this study, both per-class and overall metrics are reported for all experiments. For each class, the F1-score, Precision, and Recall are reported. At the aggregate level, the mean F1-score (mF1), mean IoU (mIoU), OA, and BOA are reported.

2.6 Loss Functions

While evaluation metrics are used to assess model performance during testing, the training process is guided by a loss function. In semantic segmentation, the loss is computed at pixel level by comparing predicted classes probabilities with reference labels [20].

The standard choice for multiclass segmentation is **categorical cross-entropy loss**, which penalizes incorrect class probabilities and encourages the model to assign a high probability to the correct class [20]:

$$\mathcal{L}_{CE} = - \sum_{c=1}^C y_c \log(\hat{y}_c), \quad (2.9)$$

where y_c is the ground truth label for class c and $\hat{y}_c$ is the predicted probability of that class.

However, cloud detection datasets are typically highly imbalanced, as the clear class usually dominates. In such cases, standard cross-entropy may lead to biased predictions, and overlap-based loss functions are often preferred [33].

Dice loss is derived from the Dice similarity coefficient, which is equivalent to the F1-score and directly optimizes the overlap between predicted and reference masks [33].

$$\mathcal{L}_{Dice} = 1 - \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (2.10)$$

In this study, Dice loss was selected as the loss function for all experiments.

3 Data and Methodology

This chapter describes the data and methodology underlying all experiments conducted in this thesis. Section 3.1 introduces the datasets used for both source and target domain training and evaluation. Section 3.2 describes the preprocessing pipeline applied to the data. Section 3.3 details the experimental setup, including the model architecture, training configuration, and experimental design. Section 3.4 describes the evaluation methodology, and Section 3.5 provides an overview of the software implementation.

3.1 Datasets

In this study, two datasets are used, each acquired by a different satellite sensor. The first consists of Sentinel-2 imagery from the CloudSEN12+ dataset, selected from several available labeled Sentinel-2 cloud detection datasets, including KappaSet [39], the Sentinel-2 Cloud Mask Catalogue [40], a cloud mask dataset developed by Hollstein et al. [9], a dataset created by Baetens and Hagolle [41], and a dataset by Luo et al. [42]. CloudSEN12+ [43] was chosen as it is presented by its authors as the largest publicly available cloud detection dataset for Sentinel-2 imagery with high-quality expert-reviewed annotations. Also, it provided the most extensive patch coverage over the target region among the evaluated options. The second dataset consists of VEN μ S imagery from a dedicated VEN μ S cloud masking dataset created by Pešek et al. [10]. Both sensors operate in the visible and near-infrared spectral range and share a sufficient number of spectrally equivalent bands to enable cross-sensor transfer learning, as described in Section 3.1.3.

An important distinction between the datasets lies in the annotation classes that are available. CloudSEN12+ provides four semantic classes: thick cloud, thin cloud, cloud shadow, and clear. The VEN μ S dataset provides three classes: thick cloud, thin cloud, and clear, with no cloud shadow annotations. The treatment of the cloud shadow class in the context of cross-sensor transfer is discussed in Section 3.3.3.

3.1.1 Sentinel-2 and CloudSEN12+

Sentinel-2 is an Earth observation mission operated by the European Space Agency (ESA) as part of the Copernicus programme. The mission includes two satellites, Sentinel-2A and Sentinel-2B, launched in June 2015 and March 2017 from French Guiana. Both satellites orbit Earth at an altitude of 786 km on a sun-synchronous polar orbit, with a 180° phase offset between them. Each satellite has a swath width of 290 km and a revisit time of 10 days. With

both satellites operating together, the revisit time is reduced to 5 days at the equator. The mission covers latitudes between 56° S and 84° N. Each satellite is equipped with the MultiSpectral Instrument (MSI), which acquires imagery in 13 spectral bands spanning the visible, near-infrared, and shortwave infrared regions (443–2190 nm) [44]. Spatial resolution varies across bands: 10 m for the three visible and one near-infrared band, 20 m for the six red-edge and shortwave infrared bands, and 60 m for the three atmospheric correction bands. The full list of spectral bands is provided in Table 3.1.

TABLE 3.1: Spectral bands of the Sentinel-2 MSI.

Source: [44]

Band	Central Wavelength (nm)	Bandwidth (nm)	Resolution (m)
B1	443	20	60
B2	490	65	10
B3	560	35	10
B4	665	30	10
B5	705	15	20
B6	740	15	20
B7	783	20	20
B8	842	115	10
B8A	865	20	20
B9	945	20	60
B10	1375	30	60
B11	1610	90	20
B12	2190	180	20

CloudSEN12+ [43] is the Sentinel-2 dataset used in this work. It is an extension of the original CloudSEN12 dataset [45] and represents the largest publicly available collection of expert-labeled pixels for cloud and cloud shadow detection in Sentinel-2 imagery. All 13 spectral bands are provided at a uniform spatial resolution of 10 m, with the native 20 m and 60 m bands upsampled accordingly. All bands are provided as Level 1C top-of-atmosphere reflectances. CloudSEN12+ provides global coverage, with pixel-wise annotations for four semantic classes: thick cloud, thin cloud, cloud shadow, and clear. Three annotation quality tiers are available: high-quality, scribble-labeled, and unlabeled.

Although CloudSEN12+ provides global coverage, only patches falling within a selected region of the Middle East and North Africa were used for this study. This geographic restriction was applied to reduce the domain shift between the source and target datasets. The region was selected based on similar land cover characteristics with the VEN μ S dataset, containing vegetated, urban, agricultural, and arid surfaces, as well as water bodies. A particular focus was placed on desert surfaces, which are spectrally challenging for cloud detection due to their high reflectance and formless nature, with no clear or systematic texture patterns. A geographically matched training set, therefore, helps the

model learn to distinguish clouds from such surfaces more reliably. The geographic distribution of the CloudSEN12+ patches within the area of interest is illustrated in Figure 3.1.



FIGURE 3.1: Geographic distribution of CloudSEN12+ patches within the area of interest.

Only the high-quality annotation subset was used to ensure the highest possible quality of annotations. CloudSEN12+ provides patches in two sizes: 509×509 pixels (p509) and 2000×2000 pixels (p2000). Since only four p2000 patches fall within the area of interest, the p2000 subset was excluded. Each p509 location is covered by five scenes with varying cloud cover fractions. The p509 subset was further filtered during preprocessing based on a visual inspection of annotation quality, as described in Section 3.2.

3.1.2 VEN μ S and the VEN μ S Cloud Masking Dataset

VEN μ S (Vegetation and Environment monitoring on a New Micro-Satellite) is a joint Earth observation mission of the French Centre National d'Études Spatiales (CNES) and the Israeli Space Agency (ISA), launched by a Vega rocket from the French Guiana Space Centre in August 2017 [46]. The satellite was developed as a demonstration mission for the GMES (Global Monitoring for Environment and Security) programme, the same initiative under which Sentinel-2 was developed, which accounts for the strong similarity in spectral band configuration between the two sensors in the visible and red-edge regions.

The mission was conducted in multiple phases, during which the orbital altitude changed, affecting the native spatial resolution of the sensor. The first phase, which is relevant for this thesis, ran from November 2017 to November 2020, during which the satellite orbited in a sun-synchronous orbit at 720 km altitude with a swath width of 27.5 km and a revisit time of two days [47].

The spectral characteristics of the sensor remained identical across all phases. The mission concluded in mid-2024.

The satellite carried the VSSC (VEN μ S SuperSpectral Camera), which provided imagery in 12 narrow spectral bands covering the visible and near-infrared range (420–910 nm) [46]. The native spatial resolution of the VSSC sensor varied depending on the orbital altitude. VEN μ S data are available at three product levels [47]. Level 1 products provide geolocated top-of-atmosphere reflectances at 5 m spatial resolution. Level 2 products provide surface reflectances, in which the atmospheric contribution to the signal is suppressed and a cloud and cloud shadow mask is generated as part of this process using the MAJA algorithm. Level 2 products are distributed at 10 m spatial resolution, as data are resampled to a coarser grid during processing. Level 3 products are ten-day composites of Level 2 data, providing cloud-free surface reflectances at regular intervals. This study uses Level 2 products exclusively. Unlike Sentinel-2, VEN μ S data do not include shortwave infrared bands, which are commonly used for cloud detection. The full list of spectral bands is provided in Table 3.2.

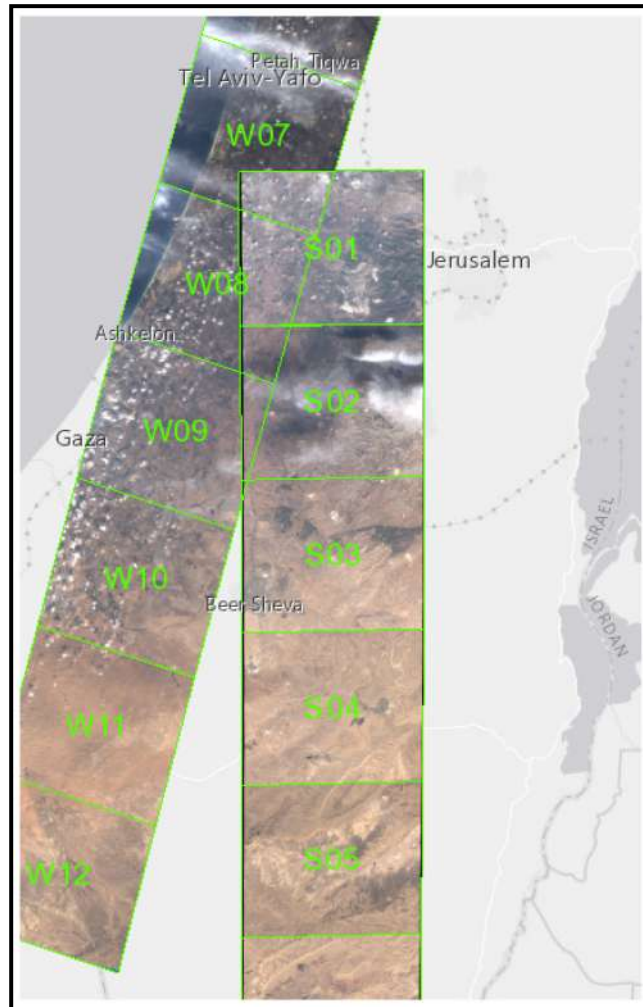
TABLE 3.2: Spectral bands of the VEN μ S VSSC.

Source: [46]

Band	Central Wavelength (nm)	Bandwidth (nm)
B1	423.9	40
B2	446.9	40
B3	491.9	40
B4	555.0	40
B5	619.7	40
B6	619.7	40
B7	666.2	30
B8	702.0	24
B9	741.1	16
B10	782.2	16
B11	861.1	40
B12	908.7	20

All bands acquired at the same spatial resolution

The VEN μ S cloud masking dataset used in this study was created by Pešek et al. [10]. The dataset covers three VEN μ S tiles over Israel: W07, S01, and S05, illustrated in Figure 3.2, selected to represent the main land cover types present in Israel, including vegetated, urban, agricultural, and arid areas as well as water bodies.

FIGURE 3.2: VEN μ S tile strips over Israel.

Source: [10]

Nine scenes were manually labeled, with more than half of the dataset consisting of scenes from the Negev desert tile S05, reflecting the particular challenge of distinguishing clouds from bright arid surfaces. The dataset was created in response to the insufficiency of automatically generated MAJA cloud masks for VEN μ S data, which were found to overestimate cloud cover, particularly over urban and arid regions [10]. Each scene is pixel-wise annotated with one of three semantic classes: clear, thick cloud, and thin cloud. No cloud shadow class is provided. All scenes are Level 2 atmospherically corrected products. Scenes from tiles W07 and S05 are provided at 10 m spatial resolution, while tile S01 is provided at 5 m spatial resolution. According to the dataset author, tile S01 also consists of Level 2 products. Given that the scene was acquired in 2018, it must originate from the first mission phase, for which 10 m resolution would be expected. As this difference in resolution does not affect the scope of this work, it was not investigated further.

3.1.3 Band Selection for Cross-Sensor Transfer

Although VEN μ S was designed as a demonstration mission for Sentinel-2 [48] and shares a similar spectral configuration in the visible and red-edge regions, it lacks shortwave infrared bands that are present in Sentinel-2. For meaningful cross-sensor transfer, only bands with spectrally equivalent counterparts in both sensors can be used. Seven such band pairs were identified by matching central wavelengths across the two sensors, as listed in Table 3.3. All selected pairs have a central wavelength difference of at most 5 nm. The selection covers the blue, green, red, three red-edge, and narrow near-infrared bands. In the CloudSEN12+ dataset, all bands are already provided at a uniform resolution of 10 m. However, the equivalent Sentinel-2 bands B5–B8A have a native resolution of 20 m and were upsampled to 10 m.

TABLE 3.3: Spectrally equivalent bands used for cross-sensor transfer learning.

VEN μ S	λ (nm)	Sentinel-2	λ (nm)	$\Delta\lambda$ (nm)	Description
B3	491.9	B2	490	1.9	Blue
B4	555.0	B3	560	5.0	Green
B7	666.2	B4	665	1.2	Red
B8	702.0	B5	705	3.0	Red-edge 1
B9	741.1	B6	740	1.1	Red-edge 2
B10	782.2	B7	783	0.8	Red-edge 3
B11	861.1	B8A	865	3.9	Narrow NIR

λ denotes the central wavelength.

$\Delta\lambda$ denotes the absolute difference in central wavelength between sensors.

3.2 Data Preprocessing

Before the datasets can be used for model training and evaluation, several preprocessing steps are required. Raw satellite imagery from different sensors differs in spectral configuration, radiometric properties, spatial resolution, and data format. These differences must be minimized to enable meaningful cross-sensor transfer. Both the Sentinel-2 and VEN μ S datasets undergo a set of preprocessing steps: spectral band extraction, annotation class remapping, dataset splitting, tile generation, and normalization. However, not all steps are applied in every experiment, as described in Section 3.3.3.

In addition to the shared steps, the two datasets require sensor-specific preprocessing. The Sentinel-2 data requires geographic filtering and filtering based on annotation quality, while the VEN μ S data requires handling of nodata values. The sensor-specific preprocessing steps are described in Sections 3.2.1 and 3.2.2 respectively, followed by a description of the common dataset generation steps in Section 3.2.3. Normalization strategies are described in Section 3.2.4.

3.2.1 Sentinel-2 Preprocessing

The initial stage of Sentinel-2 preprocessing involved applying a geographic filter. Since CloudSEN12+ provides global coverage, only patches intersecting selected countries in the Middle East and North Africa region (Libya, Egypt, Israel, Lebanon, Syria, Jordan, Iraq, and Saudi Arabia) were retained. To achieve this, the spatial extent of each patch was checked against country bounding boxes, and only patches intersecting the selected countries were retained for further processing.

In the second step, the seven spectral bands selected for cross-sensor transfer, as described in Section 3.1.3, were extracted from each patch. Since CloudSEN12+ stores the annotation mask as a separate band within the same file, the labels were extracted and saved as independent files during this step. The original four-class annotation scheme was remapped to three classes by merging the thin cloud class into the thick cloud class, resulting in cloud, clear, and cloud shadow labels. All 13 spectral bands in CloudSEN12+ are provided at a uniform spatial resolution of 10 m, so no resampling was required.

Following band extraction, a manual quality review was performed on all patches within the area of interest. Each patch was visually inspected and assigned one of three labels: good, bad, or clear. Patches were marked as bad primarily due to annotation quality issues, such as incorrect or incomplete cloud labels. Patches marked as clear contained no cloud coverage. Since cloud-free patches contribute only clear pixels, which are already well represented within partially cloudy patches, retaining all of them would increase the training data volume without providing additional value for cloud detection. Only a subset of clear patches was thus kept to maintain a representative sample of cloud-free conditions while avoiding unnecessary duplication. A total of 129 patches were excluded during this review, leaving 140 patches for further processing.

3.2.2 VEN μ S Preprocessing

Similarly to Sentinel-2 preprocessing, the first step involved extracting the seven spectral bands selected for cross-sensor transfer, as described in Section 3.1.3. The original three-class annotation scheme was remapped to two classes by merging the thin cloud class into the thick cloud class, resulting in cloud and clear labels. No cloud shadow class is present in the VEN μ S dataset.

A key difference from the Sentinel-2 preprocessing is the presence of nodata pixels in VEN μ S scenes. In the annotation masks, nodata regions were filled with the ignore value of 255, which was later used to exclude these pixels from loss computation during training, as described in Section 3.2.3.

Unlike the Sentinel-2 dataset, no manual quality review was performed for the VEN μ S scenes, as an initial inspection indicated sufficient annotation quality across all scenes.

3.2.3 Dataset Generation

Following sensor-specific preprocessing, data from both datasets went through a set of modifications in order to create training datasets to be used in the U-Net model. All data were tiled into 512×512 pixel tiles, which serve as the input for the U-Net model. Although U-Net is a fully convolutional architecture capable of processing inputs of variable spatial dimensions, the repeated pooling operations in the encoder require input dimensions to be divisible by 2^d , where d is the number of pooling layers. For a standard U-Net with five pooling layers, this means input dimensions must be divisible by 32. Since 509 is not divisible by 32, padding is required. A tile size of 512×512 pixels was selected as the smallest valid dimension larger than 509. A tile size of 256×256 pixels was also tested, but initial experiments showed that larger tiles provide better spatial context for accurate identification of cloud edges.

For CloudSEN12+ patches, which have a native size of 509×509 pixels, reflective padding of 3 pixels was applied to the right and bottom edges to reach the required 512×512 pixel dimensions. Reflective padding was chosen over zero padding to avoid introducing artificial boundaries that could affect model predictions near tile edges. For VEN μ S scenes, which have a native size of 2048×2048 pixels, tiling was performed using a non-overlapping grid.

Both datasets were divided into training, validation, and test subsets. The test set was manually selected prior to dataset generation to represent the full range of cloud types and land cover classes present in each dataset, and was excluded from all subsequent training and validation steps. For both datasets, ten tiles were selected for the test set. The remaining tiles were automatically divided into training and validation subsets using a 75/25 train/validation ratio.

During dataset generation, data augmentation was applied to the training set in the form of rotations by 90° , 180° , and 270° , effectively quadrupling the number of training tiles. Validation and test sets were not augmented, as they should represent real data conditions. Padded regions and nodata pixels were excluded from loss computation during training using the ignore value of 255 assigned to the annotation masks. The final dataset composition for both datasets is summarized in Table 3.4.

TABLE 3.4: Dataset composition for Sentinel-2 and VEN μ S after preprocessing and dataset generation.

	Sentinel-2	VEN μ S
After geographic filtering	269	—
Excluded during quality review	129	—
Available after filtering	140	135
Test set	10	10
Training set	98	93
Validation set	32	32
Training set (augmented)	392	372

Augmented training set obtained by rotations of 90° , 180° , and 270° .

Dash (—) indicates steps not applicable for VEN μ S.

3.2.4 Normalization Strategies

Normalization is an important preprocessing step in many deep learning applications [49]. It helps to reduce the influence of different value ranges in the input data, supporting faster convergence and better generalization. In cross-sensor transfer learning, normalization is particularly relevant as inputs from different sensors can have different value distributions, with values from the target sensor potentially falling outside the range seen during training [4]. Normalization statistics are always computed from the training split only, as using validation or test data would create data leakage. Four normalization strategies are investigated in this study.

The first strategy uses **raw** data without any normalization. The original reflectance values are passed directly to the model, preserving the full radiometric information but exposing the model to the full cross-sensor radiometric differences during transfer learning.

The remaining three strategies are based on Z-score normalization, which transforms each band independently according to:

$$\hat{x}_b = \frac{x_b - \mu_b}{\sigma_b} \quad (3.1)$$

where x_b is the original pixel value in band b , μ_b is the mean, and σ_b is the standard deviation. The three strategies differ in how μ_b and σ_b are computed.

In **fixed Z-score normalization with Sentinel-2 statistics**, μ_b and σ_b are computed from the Sentinel-2 training set, excluding nodata pixels. The same statistics are then applied to both the Sentinel-2 and VEN μ S data. This approach maps both datasets into the same radiometric space defined by the Sentinel-2 training distribution, which is particularly relevant for the transductive transfer setting, where the model is trained on Sentinel-2 data and evaluated on VEN μ S data without any target domain adaptation.

In **fixed Z-score normalization with VEN μ S statistics**, μ_b and σ_b are computed from the VEN μ S training set only, excluding nodata pixels, and are then used to normalize VEN μ S data. This approach maps the VEN μ S data into a radiometric space defined by the sensor-specific statistics and atmospheric correction characteristics, making it the natural choice for the inductive and from-scratch settings where the model is trained and evaluated entirely within the target domain.

In **dynamic Z-score normalization**, proposed by Wright et al. [4], μ_b and σ_b are computed independently for each individual tile from its own valid pixels, excluding nodata values. Unlike fixed normalization strategies that use dataset-wide statistics, this approach normalizes each tile independently to have zero mean and unit standard deviation per band, regardless of the sensor, processing level, or acquisition conditions. This makes it particularly suitable for cross-sensor transfer learning as it effectively eliminates radiometric differences between sensors without requiring any prior knowledge of the

target sensor’s statistics. The four normalization strategies are summarized in Table 3.5.

TABLE 3.5: Summary of normalization strategies used in this study. S2: Sentinel-2

Strategy	Statistics source	Applied to	Scope
Raw	—	S2, VEN μ S	—
Fixed Z-score (S2 statistics)	S2 training set	S2, VEN μ S	Global
Fixed Z-score (VEN μ S statistics)	VEN μ S training set	VEN μ S	Global
Dynamic Z-score	Per tile	S2, VEN μ S	Per tile

3.2.5 Preprocessing Pipeline

The preprocessing steps described in this chapter were implemented as a modular Python pipeline consisting of a shared set of scripts used for both Sentinel-2 and VEN μ S data, a master script orchestrating the full pipeline execution, and a sensor-specific JSON configuration file specifying dataset paths, band indices, class remapping, and normalization parameters. The manual quality review step is implemented as an interactive Jupyter notebook, enabling visual inspection of each patch and assignment of quality labels. The full pipeline structure is illustrated in Figure 3.3. Further implementation details are provided in Section 3.5.

3.3 Experimental Setup

This section describes the model architecture, training configuration, and experimental design used throughout the study. The model architecture and training configuration remain consistent across all experiments. The experimental design defines the full set of experiments investigated, the motivation behind them, and the normalization strategies applicable to each.

3.3.1 Model Architecture

All experiments in this study use the U-Net architecture, which is described in detail in Section 2.3.1 and selected on the basis of its consistently strong performance in cloud detection tasks. The implementation is provided by `cnn-lib`, an open-source Python library for semantic segmentation of remote sensing imagery using CNNs, developed by the thesis supervisor and further extended during this thesis, as described in Section 3.5. The implementation follows the original architecture proposed by Ronneberger et al. [26], extended with support for dropout, batch normalization, and configurable padding. The model is implemented in TensorFlow and Keras.

The model accepts 7 input channels corresponding to the spectrally equivalent bands described in Section 3.1.3. For Sentinel-2 training, the model produces a 3-class output: cloud, clear, and cloud shadow. For VEN μ S fine-tuning and

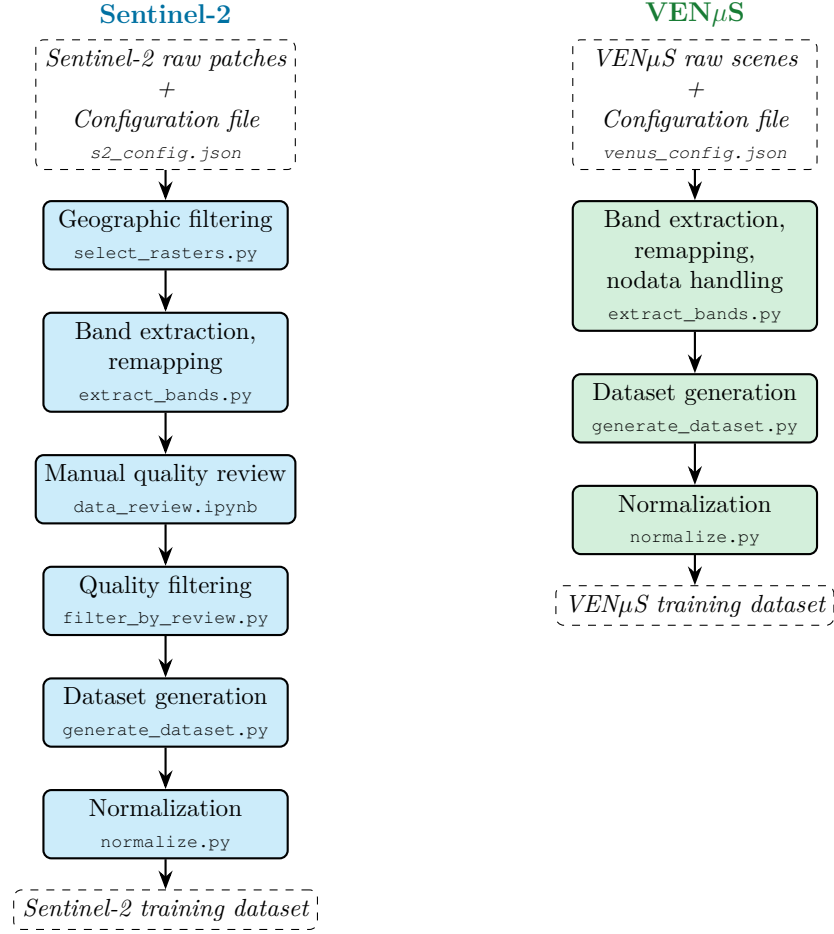


FIGURE 3.3: Structure of the preprocessing pipeline.

from-scratch experiments, the output layer is configured for 2 classes: cloud and cloud-free, since no cloud shadow annotations are available in the VENμS dataset. The model is initialized with random weights for all experiments involving training from scratch.

3.3.2 Training Configuration

All models were trained using the Adam optimizer with a learning rate of 10^{-3} . The Dice loss function was used for all experiments, which is well suited for imbalanced segmentation tasks as described in Section 2.6. A batch size of 16 tiles of 512×512 pixels was used for all experiments. Training was performed on a Linux server equipped with an NVIDIA Tesla V100 GPU with 32 GB of memory. The training was governed by early stopping, which monitors the validation loss and terminates training if no improvement is observed within a specified number of epochs. Early stopping with a patience of 200 epochs was used for from-scratch training, reflecting the slower convergence expected when training from random initialization. For fine-tuning experiments, a patience of 100 epochs was used, where faster convergence is expected due to the pretrained weights.

3.3.3 Transfer Learning Setup and Experimental Design

The experimental design is motivated by three research questions:

1. Can a U-Net model trained on Sentinel-2 data generalize to VEN μ S imagery using only spectral band alignment as a base domain adaptation, and how does additional input-level radiometric normalization affect this generalization?
2. How much of the pretrained network needs to be adapted to achieve effective cross-sensor transfer, and does fine-tuning the full network outperform partial adaptation strategies?
3. Can transfer learning from Sentinel-2 achieve performance comparable to a model trained from scratch on VEN μ S data, and if so, what degree of adaptation is required to achieve this?

Across all transfer learning experiments, the Sentinel-2 pretrained model is assumed to have been trained independently of any specific target domain. This reflects a realistic transfer learning scenario in which the source model may have been developed for a different application, or where the target sensor data are not yet available at source training time, for example, when the target satellite has not yet been launched. Training the Sentinel-2 model with VEN μ S-derived statistics would therefore violate this assumption, as it would require prior knowledge of the target domain at source training time.

To address these questions, the experiments are organized into two groups: a baseline group, which provides reference points for evaluating transfer learning performance, and a transfer learning group, which investigates transductive and inductive settings as defined in Section 2.4. Prior to any transfer learning, a U-Net model is trained on the Sentinel-2 dataset, producing a pretrained model that serves as the initialization point for all transfer learning experiments.

Baseline

Two baselines are included. MAJA [17] serves as a classical reference providing a point of comparison against the operational algorithm natively used for VEN μ S cloud masking. However, due to the known limitation of MAJA in overestimating cloud cover over urban and arid regions [10], it is evaluated visually only and is not included in quantitative comparisons. The second baseline is a U-Net model trained from scratch on VEN μ S data. In the context of transfer learning, this model represents the performance achievable when labeled target domain data are available and no transfer from the source domain is required. It thus serves as the primary deep learning reference point, indicating how close transfer learning approaches can get to a fully supervised target domain solution. Three normalization strategies are evaluated for the from-scratch model: raw, fixed Z-score with VEN μ S statistics, and dynamic Z-score. Fixed Z-score with Sentinel-2 statistics is not applicable in this setting, as no Sentinel-2 pretrained model is involved and applying Sentinel-2

dataset statistics to VEN μ S data without a corresponding pretrained model would introduce an arbitrary radiometric shift with no benefit.

Transductive Transfer Learning

In the transductive setting, the Sentinel-2 pretrained model is applied directly to VEN μ S data without any modification to the model weights, as defined in Section 2.4. This setting corresponds to the scenario where no labeled target domain data are available, and the source model must generalize directly to the target domain, addressing the first research question. In all transductive experiments, a base level of domain adaptation is achieved through the selection of spectrally equivalent bands, as described in Section 3.1.3, which reduces the feature space mismatch between the two sensors. When no normalization is applied beyond this, the experiment evaluates the extent to which spectral alignment alone is sufficient for cross-sensor transfer. When normalization is additionally applied, it constitutes a further step of input-level domain adaptation, progressively reducing the radiometric gap between sensors. The transductive setting also preserves the full 3-class output of the Sentinel-2 model, including cloud shadow, which cannot be maintained in any fine-tuning setting due to the absence of shadow annotations in the VEN μ S dataset. In evaluation, shadow predictions are collapsed into the cloud-free class, as described in Section 3.4. All four normalization strategies are evaluated in this setting. The fixed Z-score normalization with VEN μ S statistics is an exception, as no corresponding Sentinel-2 pretrained model exists for this configuration. In this case, the model pretrained using fixed Z-score normalization with Sentinel-2 statistics is used, introducing an additional source of inconsistency between training and inference normalization for this variant.

Inductive Transfer Learning

In the inductive setting, the pretrained model is adapted to VEN μ S data using labeled target domain samples, as defined in Section 2.4. This setting addresses the second and third research questions by investigating how much of the network needs to be updated for effective transfer, and whether adapted models can match the from-scratch baseline and MAJA. Three fine-tuning strategies are investigated, differing in the degree of network adaptation. All inductive experiments are initialized from the best Sentinel-2 checkpoint, defined as the epoch with the lowest validation loss during Sentinel-2 training. Since no cloud shadow annotations are available in the VEN μ S dataset, the 3-class output layer is replaced with a randomly initialized 2-class output layer in all inductive experiments, and evaluation is performed on cloud and cloud-free classes only.

The first strategy fine-tunes only the output layer, leaving all encoder and decoder weights frozen. This represents the minimal adaptation case and tests whether the feature representations learned on Sentinel-2 data are directly transferable to VEN μ S, requiring only a remapping of the class boundaries. The second strategy freezes the encoder and fine-tunes the decoder together

with the output layer. This investigates whether the encoder representations generalize sufficiently across sensors while the decoder requires adaptation to the radiometric and spatial properties of the target domain. The third strategy fine-tunes the full network by allowing all layers to adapt to the target domain. Together, these three strategies enable a systematic comparison of how much of the network needs to be updated for effective cross-sensor transfer. All four normalization strategies are evaluated for each fine-tuning variant.

The full experimental design is summarized in Table 3.6, and the normalization strategies applicable to each experiment are shown in Table 3.7.

TABLE 3.6: Summary of all experiments in this study.

ID	Name	Setting	Fine-tuned scope	Patience
B1	MAJA	Baseline	—	—
B2	From scratch	Baseline	—	200
E0	Transductive	Transductive	None	—
E1	Output layer	Inductive	Output layer	100
E2	Frozen encoder	Inductive	Decoder + output	100
E3	Full fine-tune	Inductive	Full network	100

All transfer learning experiments use the best Sentinel-2 checkpoint as initialization. E0 produces 3-class output; all other VEN μ S experiments produce 2-class output.

TABLE 3.7: Normalization strategies applied to each experiment.

Experiment	Raw	Fixed (S2)	Fixed (VEN μ S)	Dynamic
B1 (MAJA)	X	X	X	X
B2 (From scratch)	✓	X	✓	✓
E0 (Transductive)	✓	✓	✓	✓
E1 (Output layer)	✓	✓	✓	✓
E2 (Frozen encoder)	✓	✓	✓	✓
E3 (Full fine-tune)	✓	✓	✓	✓

Fixed (S2): statistics computed from Sentinel-2 training set

Fixed (VEN μ S): statistics computed from VEN μ S training set.

Dynamic: statistics computed per tile from valid pixels.

3.4 Evaluation

Model performance is evaluated on testing datasets from both sensors. The Sentinel-2 test set consists of ten patches selected during preprocessing, as described in Section 3.2.3, and is used to assess source domain performance. The VEN μ S test set consists of ten tiles and serves as the primary benchmark for evaluating cross-sensor transfer learning performance.

For all experiments, evaluation is performed on two classes: cloud and cloud-free. In the transductive setting, the Sentinel-2 trained model produces 3-class

predictions including cloud shadow. Since the VEN μ S ground truth contains no shadow annotations, shadow predictions are collapsed into the cloud-free class prior to evaluation, as described in Section 3.3.3.

Performance is measured by comparing predicted pixel labels against the ground truth annotations using a confusion matrix, from which evaluation metrics are derived, as described in Section 2.5. Multiple metrics are reported to provide a complete picture of model performance, as different metrics emphasize different aspects of segmentation quality, and different studies use different primary metrics, making direct comparison across works easier when all are reported. Both confusion matrices and evaluation metrics are reported for all experiments.

3.5 Implementation

The software developed as part of this thesis consists of three components: a preprocessing pipeline, extensions to `cnn-lib`, and evaluation and visualization scripts. The preprocessing pipeline and evaluation and visualization scripts are publicly available in the thesis repository.¹ The `cnn-lib` library is maintained in a separate repository.²

The preprocessing pipeline implements all steps described in Section 3.2 as a modular set of Python scripts and an interactive Jupyter notebook for manual quality review, configured through sensor-specific JSON files, as described in Section 3.2.5.

The `cnn-lib` library, used for model training, transfer learning, and inference throughout this study, was extended during this thesis with a padding implementation supporting multiple padding types, layer freezing by name pattern, tolerance for weight shape mismatches between source and target domains, ignore value functionality for excluding padded regions from loss computation. In addition, several bugs affecting training stability and evaluation correctness were identified and fixed.

The evaluation and visualization scripts compute per-class and overall metrics from model predictions and generate confusion matrices and visual prediction examples as reported in Chapter 4.

Claude,³ a large language model developed by Anthropic, was used as a supportive tool during the preparation of this thesis for programming assistance and code debugging, stylistic refinement, and text reformulation.

¹<https://github.com/ctu-geoforall-lab-projects/dp-klimes-2026>

²<https://github.com/ctu-geoforall-lab/cnn-lib>

³<https://www.anthropic.com>

4 Experimental Results

This chapter presents the results of all experiments defined in Section 3.3.3. The chapter is organized as follows. Section 4.1 reports the results of Sentinel-2 model training, which provides the pretrained model used as the starting point for all transfer learning experiments. Section 4.2 presents the baseline results, including the from-scratch $\text{VEN}_{\mu\text{S}}$ model and MAJA. Sections 4.3 and 4.4 report the results of the transductive and inductive transfer learning experiments. Finally, Section 4.5 summarizes the results across all experiments and addresses the research questions defined in Section 3.3.3.

For each experiment, per-class and overall metrics for all normalization strategies are reported in the main text, while confusion matrices and visual examples of predicted cloud masks are shown for the best performing normalization strategy only. Additional visual prediction examples for the best performing normalization strategy in each experiment are provided in Appendix A.

4.1 Sentinel-2 Model Training

Before transfer learning experiments can be conducted, a U-Net model is trained on the Sentinel-2 dataset for each applicable normalization strategy. Fixed Z-score with $\text{VEN}_{\mu\text{S}}$ statistics is not applicable here, as the model is trained entirely on Sentinel-2 data and $\text{VEN}_{\mu\text{S}}$ statistics have no meaningful interpretation in this context. The resulting three models serve as the initialization point for all subsequent transfer learning experiments. The Sentinel-2 test set metrics are reported here as a reference for source domain performance.

The per-class and overall metrics for all three Sentinel-2 models on the Sentinel-2 test set are summarized in Tables 4.1 and 4.2.

TABLE 4.1: Per-class metrics for Sentinel-2 models on the Sentinel-2 test set. Norm.: Normalization strategy. Prec.: Precision. Rec.: Recall. S2: Sentinel-2.

Norm.	Cloud			Clear			Shadow		
	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.
Raw	0.746	0.844	0.668	0.789	0.815	0.764	0.534	0.417	0.744
Fixed (S2)	0.777	0.741	0.817	0.815	0.868	0.768	0.706	0.663	0.756
Dynamic	0.884	0.911	0.859	0.920	0.895	0.948	0.791	0.818	0.766

TABLE 4.2: Overall metrics and training convergence for Sentinel-2 models. Norm.: Normalization strategy. S2: Sentinel-2. Dice loss: Dice loss computed over the validation split at the best epoch.

Norm.	mF1	mIoU	OA	BOA	Best epoch	Dice loss
Raw	0.690	0.537	0.726	0.726	294	0.296
Fixed (S2)	0.766	0.623	0.784	0.780	798	0.278
Dynamic	0.865	0.766	0.890	0.858	521	0.246

The dynamic Z-score normalization produced substantially better results than both raw and fixed Z-score strategies across all metrics and all classes. The dynamic model achieved an mF1 of 0.865 and mIoU of 0.766, compared to 0.766 and 0.623 for fixed Z-score and 0.690 and 0.537 for raw data. This improvement can likely be explained by the per-tile normalization, which reduces radiometric differences between patches captured under varying atmospheric conditions and at different geographic locations, making the problem more consistent across the entire dataset.

Shadow detection shows the largest performance gap between normalization strategies. Dynamic Z-score achieves an F1 of 0.791 for shadow, compared to 0.706 for fixed Z-score and 0.534 for raw data. Shadow is consistently the hardest class across all normalizations. In the raw strategy, shadow recall is relatively high at 0.744 while precision is substantially lower at 0.417, suggesting that the model detects shadow regions, but produces predictions that extend beyond the annotated boundaries. This behaviour is likely caused by the spectral similarity of cloud shadows to other dark surfaces, compounded by the fact that shadow annotations were the most difficult to label accurately. This gap narrows with normalization, with dynamic Z-score achieving a more balanced precision of 0.818 and recall of 0.766.

The confusion matrix for the dynamic Z-score model, shown in Figure 4.1, further illustrates these patterns. The model performs well on the clear and cloud classes, with recall values of 0.95 and 0.86. Shadow shows the weakest recall at 0.77, with 10% of true shadow pixels misclassified as clear and 13% as cloud. The cloud class shows some confusion with clear (11%), which is expected at cloud boundaries, where the difference between these two classes is hard to distinguish.

The fixed Z-score model required significantly more epochs to converge (798) compared to raw (294) and dynamic (521), suggesting that global statistics introduce more radiometric variation across patches than per-tile normalization.

Representative examples of Sentinel-2 model predictions on the test set are shown in Figure 4.2. The figure illustrates the model’s ability to detect both clouds and cloud shadows over multiple land cover types.

True label	clear [1281194]	0.95	0.03	0.03
	cloud [955299]	0.11	0.86	0.03
	shadow [354317]	0.10	0.13	0.77
		clear	cloud	shadow
		Predicted label		

FIGURE 4.1: Confusion matrix for the Sentinel-2 model with dynamic Z-score normalization on the Sentinel-2 test set. Values are normalized by row.

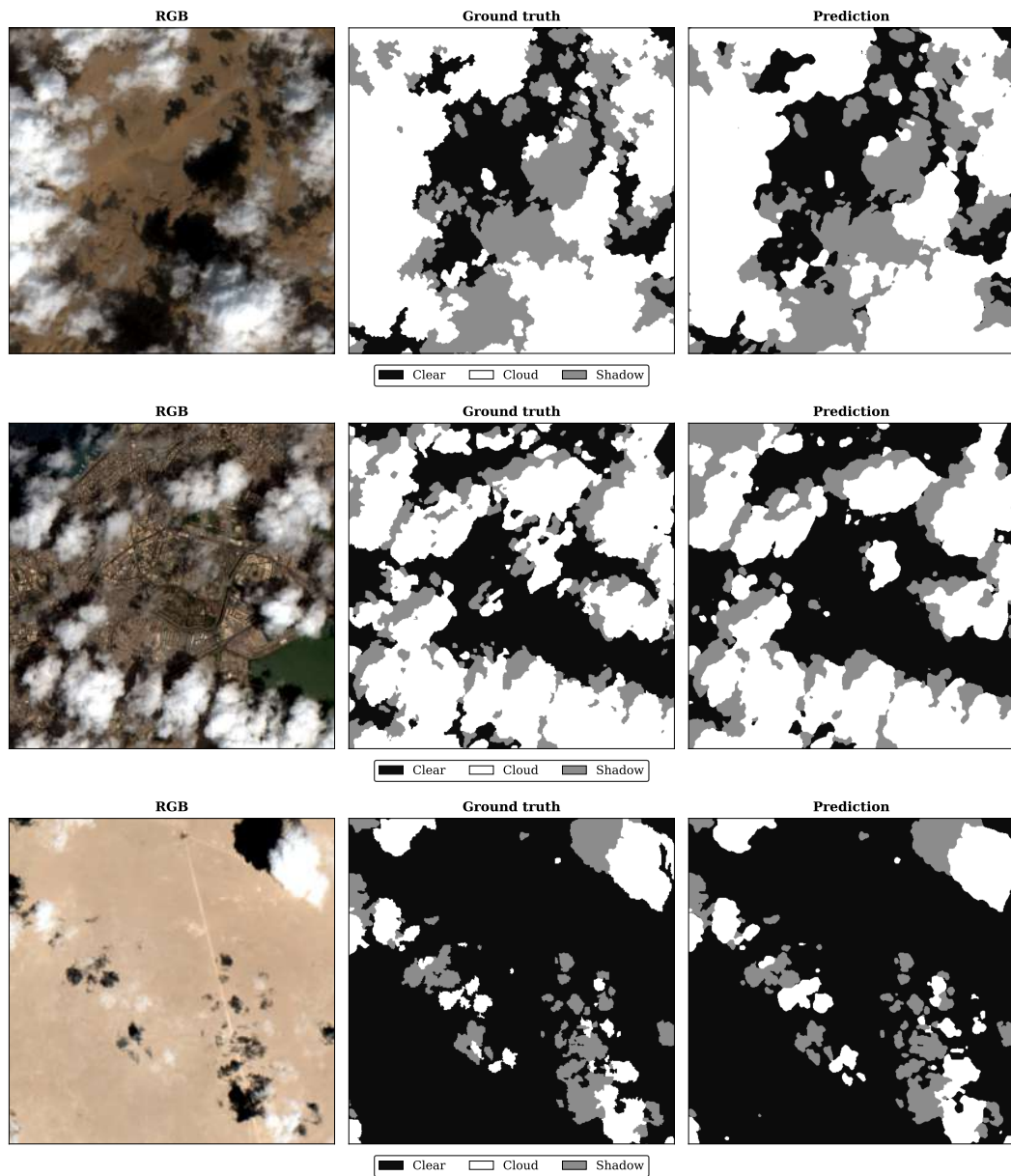


FIGURE 4.2: Representative Sentinel-2 test set predictions using dynamic Z-score normalization.

4.2 Baseline Results

This section presents the results of the two baseline experiments: a U-Net model trained from scratch on VEN μ S data and MAJA. These baselines serve as reference points for evaluating the transfer learning experiments in the following sections.

4.2.1 VEN μ S from Scratch

The per-class and overall metrics for all three from-scratch models on the VEN μ S test set are summarized in Tables 4.3 and 4.4.

TABLE 4.3: Per-class metrics for the VEN μ S from-scratch model on the VEN μ S test set. Norm.: Normalization strategy. Prec.: Precision. Rec.: Recall.

Norm.	Cloud			Cloud-free		
	F1	Prec.	Rec.	F1	Prec.	Rec.
Raw	0.926	0.952	0.902	0.982	0.976	0.989
Fixed (VEN μ S)	0.920	0.979	0.867	0.981	0.967	0.995
Dynamic	0.900	0.973	0.838	0.977	0.960	0.994

TABLE 4.4: Overall metrics and training convergence for the VEN μ S from-scratch model on the VEN μ S test set. Norm.: Normalization strategy. Dice loss: Dice loss computed over the validation split at the best epoch.

Norm.	mF1	mIoU	OA	BOA	Best epoch	Dice loss
Raw	0.954	0.914	0.971	0.945	538	0.268
Fixed (VEN μ S)	0.950	0.907	0.969	0.931	596	0.260
Dynamic	0.938	0.886	0.962	0.916	283	0.267

The from-scratch model achieves strong performance across all normalization strategies, with mF1 values ranging from 0.938 to 0.954. Unlike the Sentinel-2 model results, the raw strategy achieves the best overall performance, followed closely by fixed Z-score with VEN μ S statistics and dynamic Z-score. This behaviour is likely caused by the geographic concentration of the VEN μ S dataset. As all scenes cover a small, spectrally consistent region of Israel, the radiometric properties are similar across tiles. In this setting, per-tile normalization likely removes useful spectral information rather than harmful radiometric variation, which explains why dynamic Z-score underperforms relative to raw data. Despite achieving the lowest overall metrics, the dynamic Z-score model converges significantly faster at epoch 283, compared to 538 for raw and 596 for fixed Z-score. The differences between normalization strategies are small overall, with a gap of only 1.6 percentage points in mF1 between raw and

dynamic, suggesting that for a geographically concentrated dataset with consistent radiometric properties, normalization strategy has less impact than for a dataset covering more diverse geographic regions and surface types.

The cloud-free class consistently outperforms the cloud class across all normalizations, which is consistent with findings reported in the cloud detection literature [10, 4].

The confusion matrix for the best performing strategy (raw) is shown in Figure 4.3.

True label	cloud-free [1942766]	0.99	0.01
	cloud [491794]	0.10	0.90
		cloud-free	cloud
		Predicted label	

FIGURE 4.3: Confusion matrix for the VEN μ S from-scratch model with raw normalization on the VEN μ S test set. Values are normalized by row.

Representative visual examples of predictions of the from-scratch VEN μ S model without normalization are shown in Figure 4.4. The figure illustrates the model’s ability to detect clouds across diverse land cover types present in the VEN μ S dataset, including vegetated, urban, agricultural, and arid surfaces, as well as water bodies.

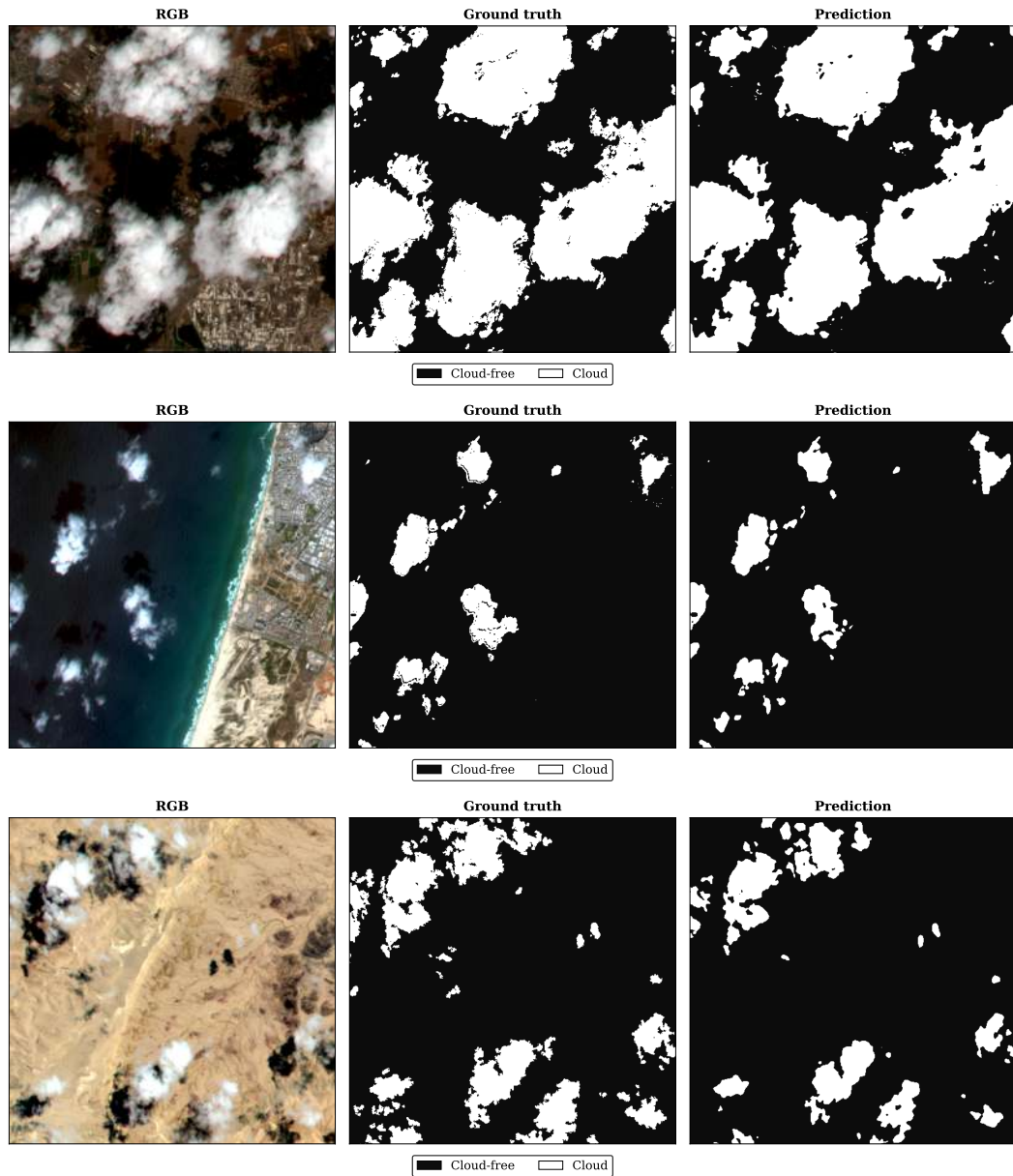


FIGURE 4.4: Representative $\text{VEN}\mu\text{S}$ from-scratch test set predictions without normalization.

4.2.2 MAJA

Due to the known limitation of MAJA in overestimating cloud cover over urban and arid regions [10], a quantitative evaluation was not performed, as discussed in Section 3.3.3. Figure 4.5 shows a comparison of MAJA masks and ground truth annotations for the three $\text{VEN}\mu\text{S}$ tiles covered in the dataset (W07, S01, and S05), clearly illustrating the systematic cloud overestimation. Given that the from-scratch model achieves an mF1 of 0.954 on the test set, deep learning approaches clearly outperform MAJA for $\text{VEN}\mu\text{S}$ cloud detection.

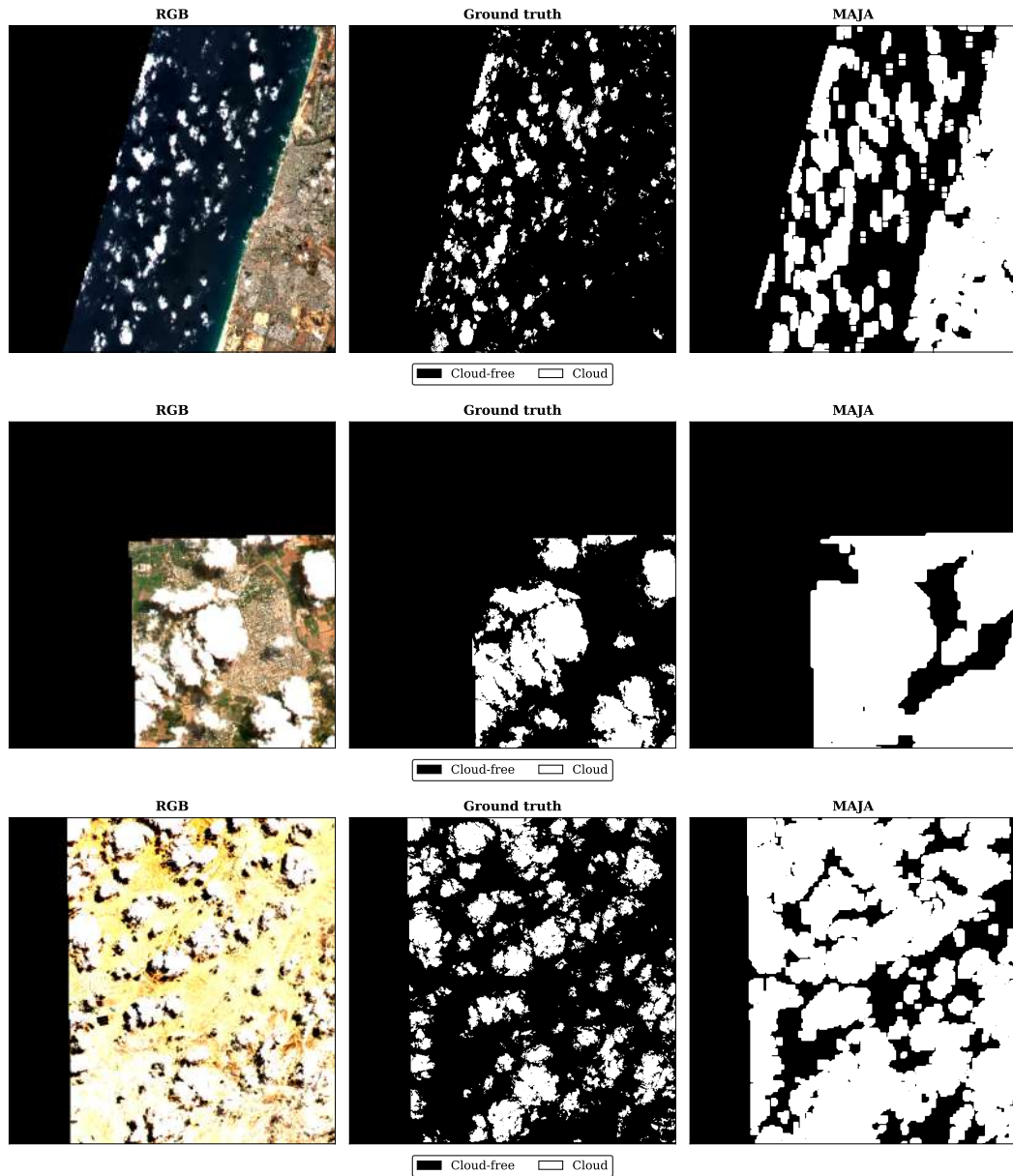


FIGURE 4.5: Comparison of MAJA cloud masks and ground truth annotations for the three VEN μ S tiles covered in the dataset.

4.3 Transductive Transfer Learning

This section presents the results of the transductive transfer learning experiment (E0), in which the Sentinel-2 pretrained model is applied directly to VEN μ S data without any modification to the model weights, addressing the first research question defined in Section 3.3.3. Each normalization variant uses the Sentinel-2 pretrained model trained with the corresponding normalization strategy. An exception is the fixed Z-score with VEN μ S statistics variant, for which no corresponding Sentinel-2 pretrained model exists. In this case, the

model pretrained with fixed Z-score with Sentinel-2 statistics is used, as it represents the closest available approximation in terms of normalization approach. Since the pretrained Sentinel-2 model predicts three classes, including cloud shadow, the transductive setting also preserves shadow predictions on VEN μ S data. As the VEN μ S dataset does not contain cloud shadow annotations, this capability cannot be evaluated quantitatively and is assessed through visual inspection only.

The per-class and overall metrics for all four normalization strategies on the VEN μ S test set are summarized in Tables 4.5 and 4.6.

TABLE 4.5: Per-class metrics for the transductive setting on the VEN μ S test set. Norm.: Normalization strategy. Prec.: Precision. Rec.: Recall. S2: Sentinel-2.

Norm.	Cloud			Cloud-free		
	F1	Prec.	Rec.	F1	Prec.	Rec.
Raw	0.005	0.018	0.003	0.868	0.792	0.960
Fixed (S2)	0.000	0.000	0.000	0.888	0.798	1.000
Fixed (VEN μ S)	0.659	0.500	0.965	0.857	0.989	0.756
Dynamic	0.874	0.793	0.973	0.963	0.993	0.936

TABLE 4.6: Overall metrics for the transductive setting on the VEN μ S test set. Norm.: Normalization strategy. S2: Sentinel-2.

Norm.	mF1	mIoU	OA	BOA
Raw	0.436	0.385	0.767	0.482
Fixed (S2)	0.444	0.399	0.798	0.500
Fixed (VEN μ S)	0.758	0.620	0.798	0.861
Dynamic	0.919	0.853	0.943	0.954

The results reveal a significant dependence on normalization strategy, with performance ranging from near-complete failure to strong cross-sensor generalization. The raw and fixed Z-score with Sentinel-2 statistics strategies both fail entirely at cloud detection, achieving cloud F1 scores of 0.005 and 0.000. Despite this, both strategies achieve relatively high overall accuracy of 0.767 and 0.798, which is a direct consequence of the class imbalance in the test set, where cloud-free pixels dominate. The model under these strategies predicts everything as cloud-free, which is reflected in the cloud-free recall values of 0.960 and 1.000. This illustrates why OA alone is a misleading metric for imbalanced datasets and highlights the importance of reporting other types of metrics.

The failure of the raw strategy confirms that spectral band alignment alone is insufficient for cross-sensor transfer when the radiometric distributions of the

two sensors differ. A key source of this difference is that the Sentinel-2 training data consists of Level 1C top-of-atmosphere reflectances, while VEN μ S Level 2 products provide atmospherically corrected surface reflectances, resulting in systematically different value ranges across sensors. The failure of the fixed Z-score with Sentinel-2 statistics strategy is consistent with this difference, as VEN μ S values fall far outside the normalized range defined by the Sentinel-2 training distribution.

Fixed Z-score normalization with VEN μ S statistics achieves a substantially better cloud F1 of 0.659 and mF1 of 0.758, demonstrating that normalizing the target data closer to its own distribution enables meaningful cloud detection. However, the results reveal a precision-recall imbalance for the cloud class, with a recall of 0.965 but precision of only 0.500, which indicates that the model detects most cloud pixels but produces a large number of false positives. The corresponding cloud-free recall drops to 0.756, suggesting significant over detection of clouds. This behaviour is consistent with the residual domain shift between the sensors. While Z-score normalization aligns the mean and standard deviation of the distributions, it does not account for differences in histogram shape, which may arise from the distinct spectral response functions of the two sensors and atmospheric correction procedures of the two datasets. As a consequence, normalized VEN μ S values fall in regions of the feature space that the model associates with clouds. It should additionally be noted that this variant uses the model pretrained with fixed Z-score with Sentinel-2 statistics, as discussed in Section 3.3.3. This introduces an additional inconsistency between training and inference normalization that likely contributes to the observed over detection.

Dynamic Z-score normalization achieves by far the strongest performance, with an mF1 of 0.919 and mIoU of 0.853. The cloud class achieves an F1 of 0.874 with a well-balanced precision of 0.793 and recall of 0.973, while the cloud-free class achieves an F1 of 0.963. This result demonstrates that per-tile normalization, by standardizing each tile independently from its own pixel statistics, effectively compensates for the radiometric gap between sensors regardless of the absolute value ranges or distribution shapes involved. Notably, the dynamic Z-score result of mF1 0.919 approaches the from-scratch baseline of 0.954, suggesting that in the absence of labeled target domain data, a well-chosen normalization strategy alone can achieve strong cross-sensor transfer performance.

The confusion matrix for the best performing strategy (dynamic Z-score) is shown in Figure 4.6. Representative visual examples of predictions under dynamic Z-score normalization are shown in Figure 4.7. The figure illustrates the model’s ability to detect clouds under dynamic Z-score normalization across vegetated, urban, agricultural, and arid surfaces, as well as water bodies.

True label	cloud-free [1942766]	0.94	0.06
	cloud [491794]	0.03	0.97
		cloud-free	cloud
		Predicted label	

FIGURE 4.6: Confusion matrix for the transductive setting (E0) with dynamic Z-score normalization on the VEN μ S test set. Values are normalized by row.

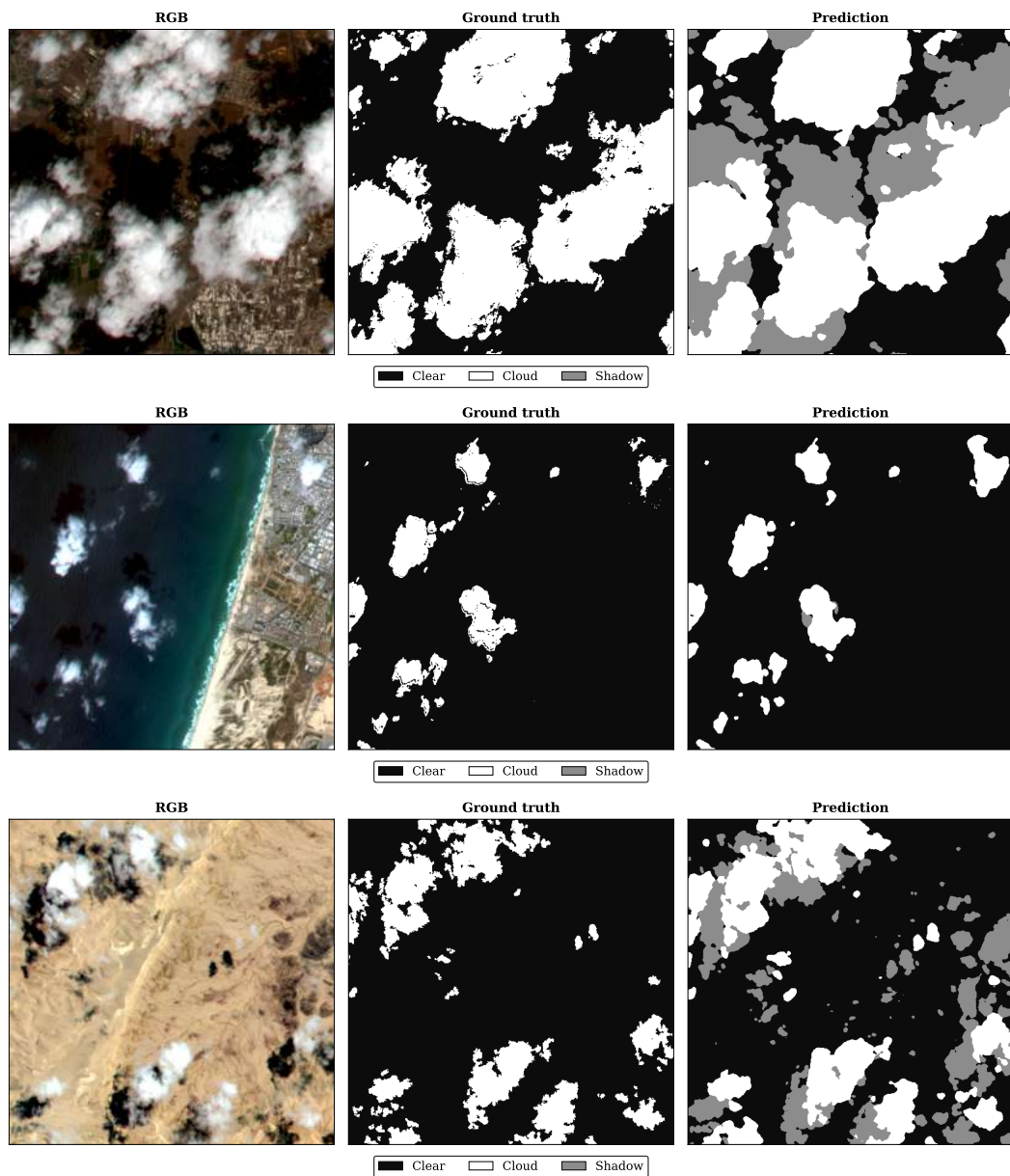


FIGURE 4.7: Representative predictions for the transductive setting (E0) with dynamic Z-score normalization on the VEN μ S test set.

4.4 Inductive Transfer Learning

This section presents the results of the three inductive transfer learning experiments: output layer fine-tuning (E1), frozen encoder fine-tuning (E2), and full fine-tuning (E3), addressing the second and third research questions defined in Section 3.3.3. All three experiments are initialized from the best Sentinel-2 checkpoint and produce 2-class outputs evaluated on cloud and cloud-free classes.

4.4.1 Output Layer Fine-Tuning

In this experiment, only the output layer is fine-tuned while all encoder and decoder weights remain frozen. The per-class and overall metrics for all four normalization strategies are summarized in Tables 4.7 and 4.8.

TABLE 4.7: Per-class metrics for output layer fine-tuning (E1) on the VEN μ S test set. Norm.: Normalization strategy. Prec.: Precision. Rec.: Recall. S2: Sentinel-2.

Norm.	Cloud			Cloud-free		
	F1	Prec.	Rec.	F1	Prec.	Rec.
Raw	0.874	0.978	0.790	0.972	0.949	0.996
Fixed (S2)	0.900	0.969	0.840	0.977	0.961	0.993
Fixed (VEN μ S)	0.922	0.976	0.874	0.982	0.969	0.994
Dynamic	0.913	0.933	0.894	0.979	0.974	0.984

TABLE 4.8: Overall metrics and training convergence for output layer fine-tuning (E1) on the VEN μ S test set. Norm.: Normalization strategy. S2: Sentinel-2. Dice loss: Dice loss computed over the validation split at the best epoch.

Norm.	mF1	mIoU	OA	BOA	Best epoch	Dice loss
Raw	0.923	0.861	0.954	0.893	212	0.300
Fixed (S2)	0.938	0.886	0.962	0.916	142	0.282
Fixed (VEN μ S)	0.952	0.910	0.970	0.934	263	0.265
Dynamic	0.946	0.899	0.966	0.939	49	0.297

Output layer fine-tuning achieves meaningful performance across all normalization strategies, with mF1 values ranging from 0.923 to 0.952. Fixed Z-score with VEN μ S statistics produces the best overall result with an mF1 of 0.952 and mIoU of 0.910, followed closely by dynamic Z-score at 0.946. Raw normalization produces the weakest results, with an mF1 of 0.923, while fixed Z-score with Sentinel-2 statistics achieves 0.938. The observation that fixed VEN μ S statistics outperform the dynamic Z-score contrasts with the transductive scenario, in which the dynamic Z-score was clearly superior. However,

the difference between the two strategies in E1 is small, at 0.006 mF1, and no strong conclusions can be drawn from this difference.

The strong performance achieved by output layer fine-tuning alone suggests that the feature representations learned by the encoder and decoder on Sentinel-2 data are largely transferable to the VEN μ S domain. This is consistent with the spectral similarity between the two sensors in the shared bands, which means the internal representations of cloud-related spectral patterns are likely to remain meaningful across these sensors, requiring only a remapping of the final class boundaries to adapt to the target domain.

A notable pattern is the high precision and lower recall for the cloud class under raw and fixed Z-score normalization with Sentinel-2 statistics strategy, suggesting that the model is conservative in its cloud predictions under these strategies, identifying cloud pixels confidently but missing a larger proportion of them. Dynamic Z-score shows a more balanced precision and recall for the cloud class at 0.933 and 0.894, and also converges significantly faster at epoch 49 compared to 212 for raw, 142 for fixed Z-score normalization with Sentinel-2 statistics, and 263 for fixed Z-score normalization with VEN μ S statistics strategy.

Comparing output layer fine-tuning against the transductive setting, output layer fine-tuning substantially improves performance under raw and fixed Sentinel-2 strategy. That demonstrates that even minimal target domain adaptation has a large impact when the normalization strategy is poorly suited for cross-sensor transfer. Under dynamic Z-score normalization, the improvement is smaller, from 0.919 to 0.946, consistent with the fact that dynamic Z-score already achieved strong performance in the transductive setting.

The confusion matrix for the best performing strategy (fixed Z-score with VEN μ S statistics) is shown in Figure 4.8. Representative visual examples of predictions under fixed Z-score normalization with VEN μ S statistics are shown in Figure 4.9.

True label	cloud-free [1942766]	0.99	0.01
	cloud [491794]	0.13	0.87
		cloud-free	cloud
		Predicted label	

FIGURE 4.8: Confusion matrix for output layer fine-tuning (E1) using fixed Z-score normalization with VEN μ S statistics on the VEN μ S test set. Values are normalized by row.

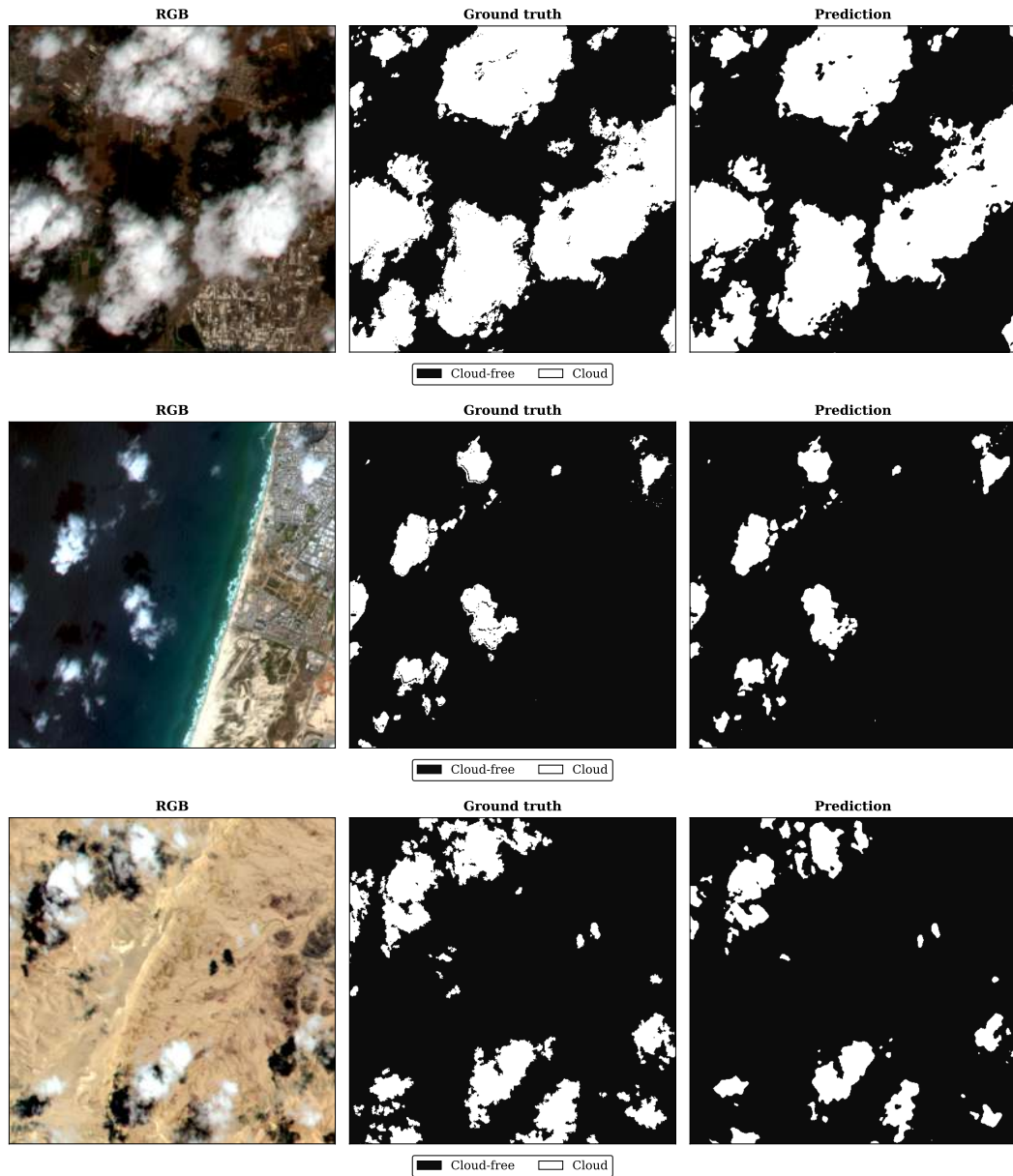


FIGURE 4.9: Representative predictions for output layer fine-tuning (E1) using fixed Z-score normalization with $\text{VEN}_{\mu\text{S}}$ statistics on the $\text{VEN}_{\mu\text{S}}$ test set.

4.4.2 Frozen Encoder Fine-Tuning

In this experiment, the encoder weights are frozen while the decoder and output layer are fine-tuned. The per-class and overall metrics for all four normalization strategies are summarized in Tables 4.9 and 4.10.

TABLE 4.9: Per-class metrics for frozen encoder fine-tuning (E2) on the VEN μ S test set. Norm.: Normalization strategy. Prec.: Precision. Rec.: Recall. S2: Sentinel-2.

Norm.	Cloud			Cloud-free		
	F1	Prec.	Rec.	F1	Prec.	Rec.
Raw	0.898	0.937	0.862	0.975	0.966	0.985
Fixed (S2)	0.880	0.980	0.798	0.973	0.951	0.996
Fixed (VEN μ S)	0.938	0.963	0.913	0.985	0.978	0.991
Dynamic	0.917	0.973	0.868	0.980	0.967	0.994

TABLE 4.10: Overall metrics and training convergence for frozen encoder fine-tuning (E2) on the VEN μ S test set. Norm.: Normalization strategy. S2: Sentinel-2. Dice loss: Dice loss computed over the validation split at the best epoch.

Norm.	mF1	mIoU	OA	BOA	Best epoch	Dice loss
Raw	0.936	0.883	0.960	0.923	412	0.284
Fixed (S2)	0.927	0.867	0.956	0.897	41	0.288
Fixed (VEN μ S)	0.961	0.926	0.975	0.952	229	0.256
Dynamic	0.949	0.904	0.968	0.931	267	0.284

Frozen encoder fine-tuning achieves stronger overall performance than output layer fine-tuning, with mF1 values ranging from 0.927 to 0.961. Fixed Z-score with VEN μ S statistics again produces the best result with an mF1 of 0.961 and mIoU of 0.926, followed by dynamic Z-score at 0.949. The improvement over E1 is consistent across normalization strategies, demonstrating that fine-tuning the decoder in addition to the output layer provides meaningful upgrade. This suggests that while the encoder representations learned on Sentinel-2 data are sufficiently general to transfer across sensors, the decoder benefits from adaptation to the spatial and radiometric properties of the VEN μ S domain.

An interesting exception is fixed Z-score normalization with Sentinel-2 statistics, which achieves a lower mF1 of 0.927 in E2 compared to 0.938 in E1, and converges very quickly at epoch 41. This unexpected result suggests that when the input normalization is poorly matched to the target domain, fine-tuning more of the network does not necessarily lead to improvement and may in fact cause the model to overfit to the training data more quickly.

The confusion matrix for the best performing strategy (fixed Z-score with VEN μ S statistics) is shown in Figure 4.10. Representative visual examples of predictions using fixed Z-score normalization with VEN μ S statistics are shown in Figure 4.11.

True label	cloud-free [1942766]	0.99	0.01
	cloud [491794]	0.09	0.91
		cloud-free	cloud
		Predicted label	

FIGURE 4.10: Confusion matrix for frozen encoder fine-tuning (E2) using fixed Z-score normalization with $\text{VEN}_{\mu\text{S}}$ statistics on the $\text{VEN}_{\mu\text{S}}$ test set. Values are normalized by row.

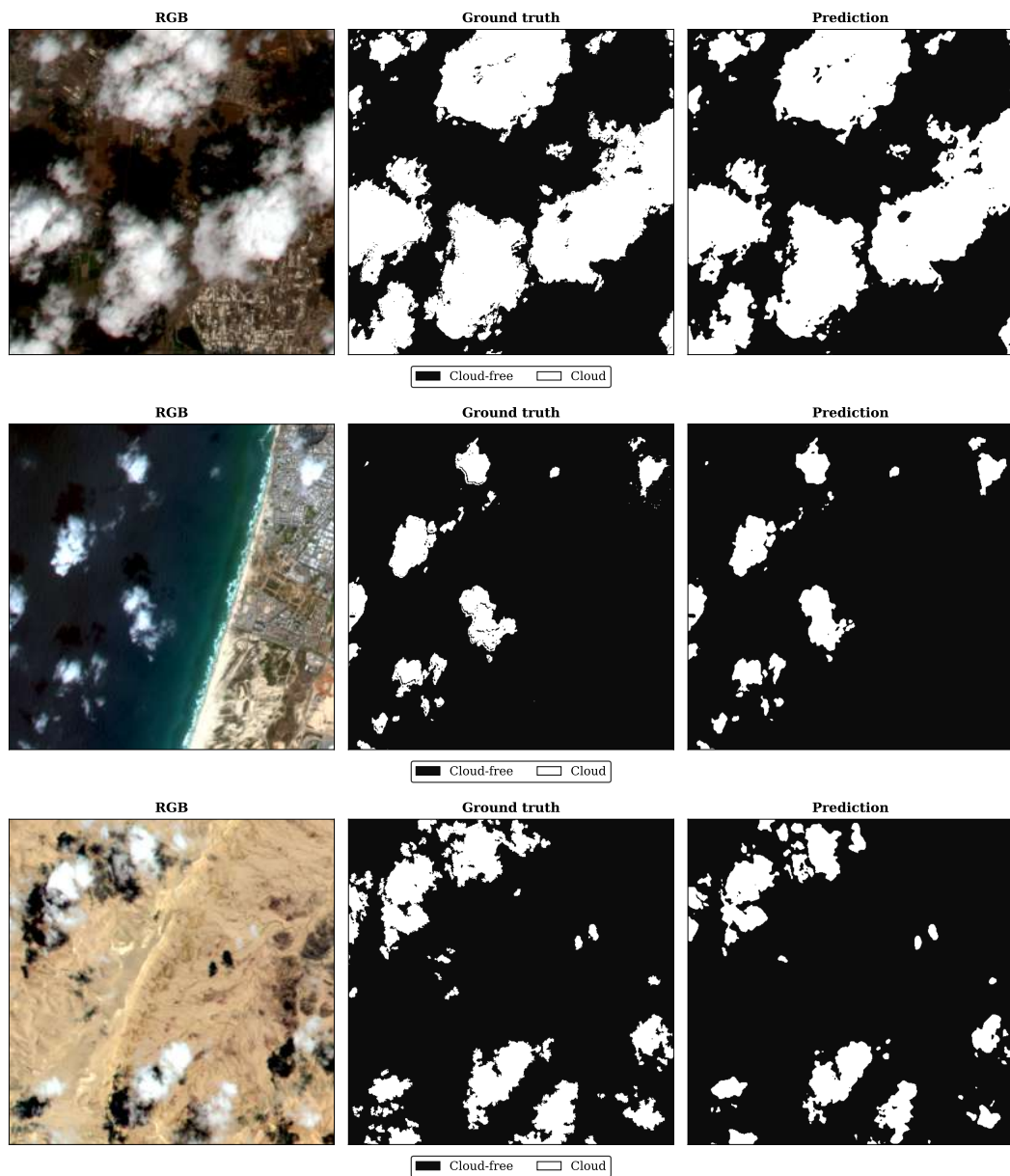


FIGURE 4.11: Representative predictions for frozen encoder fine-tuning (E2) using fixed Z-score normalization with $\text{VEN}_{\mu\text{S}}$ statistics on the $\text{VEN}_{\mu\text{S}}$ test set.

4.4.3 Full Fine-Tuning

In this experiment, all network weights including the encoder, decoder, and output layer are fine-tuned on the VEN μ S training data. The per-class and overall metrics for all four normalization strategies are summarized in Tables 4.11 and 4.12.

TABLE 4.11: Per-class metrics for full fine-tuning (E3) on the VEN μ S test set. Norm.: Normalization strategy. Prec.: Precision. Rec.: Recall. S2: Sentinel-2.

Norm.	Cloud			Cloud-free		
	F1	Prec.	Rec.	F1	Prec.	Rec.
Raw	0.891	0.965	0.827	0.975	0.958	0.992
Fixed (S2)	0.901	0.973	0.839	0.977	0.961	0.994
Fixed (VEN μ S)	0.928	0.969	0.890	0.983	0.973	0.993
Dynamic	0.928	0.969	0.891	0.983	0.973	0.993

TABLE 4.12: Overall metrics and training convergence for full fine-tuning (E3) on the VEN μ S test set. Norm.: Normalization strategy. S2: Sentinel-2. Dice loss: Dice loss computed over the validation split at the best epoch.

Norm.	mF1	mIoU	OA	BOA	Best epoch	Dice loss
Raw	0.933	0.877	0.959	0.910	106	0.284
Fixed (S2)	0.939	0.888	0.963	0.917	107	0.278
Fixed (VEN μ S)	0.955	0.915	0.972	0.941	229	0.259
Dynamic	0.956	0.916	0.972	0.942	51	0.270

Full fine-tuning achieves competitive performance across all normalization strategies, with mF1 values ranging from 0.933 to 0.956. Dynamic Z-score produces the marginally best result with an mF1 of 0.956, followed very closely by fixed Z-score with VEN μ S statistics at mF1 of 0.955. The difference between these two strategies is negligible, suggesting that when the full network is free to adapt, the choice between these two normalization strategies has little impact on final performance.

Notably, full fine-tuning does not consistently outperform frozen encoder fine-tuning. E2 normalized using fixed VEN μ S statistics achieves an mF1 of 0.961, which is higher than the best E3 result of 0.956. This is a counter-intuitive finding, as allowing more parameters to be updated might be expected to yield better results. A possible explanation is that the encoder representations learned on Sentinel-2 data are already well suited for cloud detection in the shared spectral bands, and allowing them to be updated during fine-tuning introduces unnecessary variation that slightly degrades the generalizable features preserved from source domain training.

Raw and fixed Sentinel-2 strategy show a similar pattern to E2, with E3 achieving slightly better results than E1 under these strategies but still underperforming relative to fixed VEN μ S and dynamic Z-score. Dynamic Z-score converges significantly faster under full fine-tuning, reaching the best epoch at 51 compared to 229 for fixed VEN μ S, 107 for fixed Sentinel-2, and 106 for raw.

The confusion matrix for the best performing strategy (dynamic Z-score) is shown in Figure 4.12. Representative visual examples of predictions under dynamic Z-score normalization are shown in Figure 4.13.

True label	cloud-free [1942766]	0.99	0.01
	cloud [491794]	0.11	0.89
		cloud-free	cloud
		Predicted label	

FIGURE 4.12: Confusion matrix for full fine-tuning (E3) using dynamic Z-score normalization on the VEN μ S test set. Values are normalized by row.

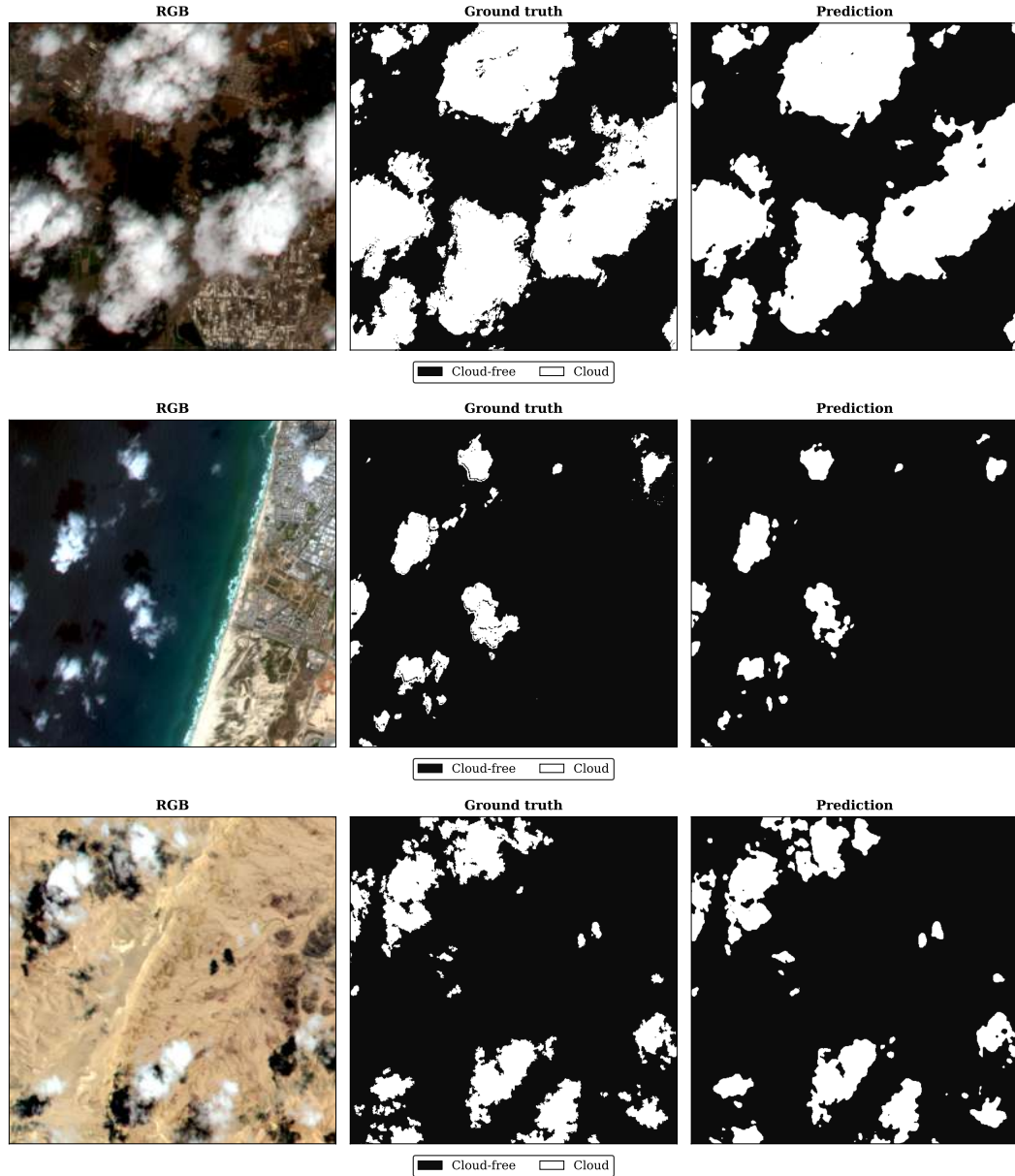


FIGURE 4.13: Representative predictions for full fine-tuning (E3) using dynamic Z-score normalization on the VEN μ S test set.

4.5 Summary of Results

This section summarizes the results across all experiments and addresses the three research questions defined in Section 3.3.3. Table 4.13 summarizes the best results achieved in each experiment, and Figure 4.14 shows the mF1 scores across all experiments and normalization strategies.

TABLE 4.13: Summary of best results across all experiments on the VEN μ S test set. For each experiment, the normalization strategy achieving the highest mF1 is reported.

Exp.	Best norm.	mF1	mIoU	OA	BOA
B2	Raw	0.954	0.914	0.971	0.945
E0	Dynamic	0.919	0.853	0.943	0.954
E1	Fixed (VEN μ S)	0.952	0.910	0.970	0.934
E2	Fixed (VEN μ S)	0.961	0.926	0.975	0.952
E3	Dynamic	0.956	0.916	0.972	0.942

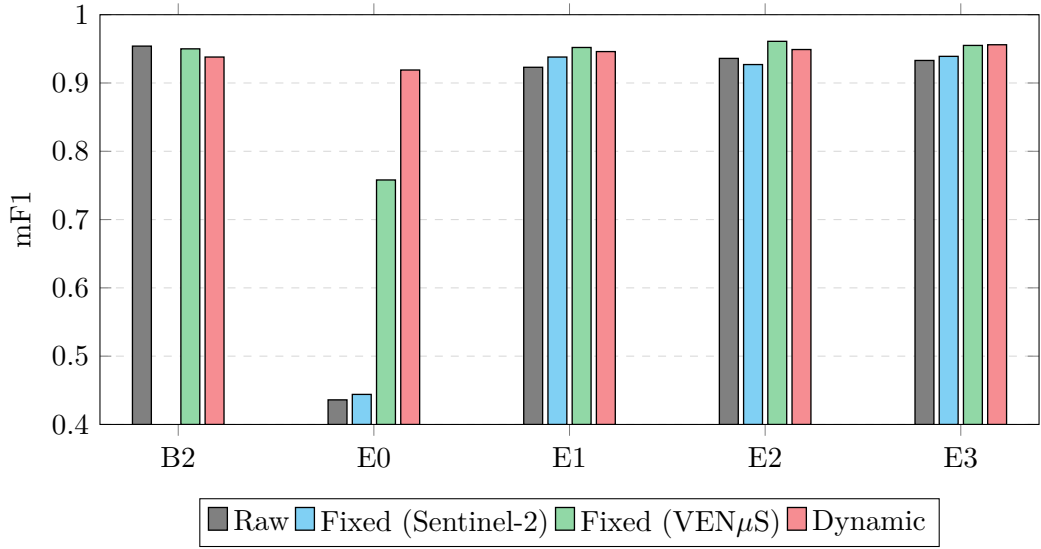


FIGURE 4.14: mF1 scores across all experiments and normalization strategies on the VEN μ S test set. Fixed (Sentinel-2) is not applicable to B2 and is therefore not shown for that experiment.

Addressing the first research question, the transductive experiment demonstrates that spectral band alignment alone is insufficient for effective cross-sensor transfer, with raw normalization achieving an mF1 of only 0.436. The large radiometric gap between the two sensors is largely attributable to the difference in product level. The Sentinel-2 training data consist of Level 1C top-of-atmosphere reflectances, while VEN μ S Level 2 products provide atmospherically corrected surface reflectances, resulting in substantially different value ranges. However, with dynamic Z-score normalization, the transductive model achieves an mF1 of 0.919, demonstrating that input-level radiometric normalization has a decisive impact on zero-shot cross-sensor generalization. This result shows that a well-chosen normalization strategy can largely bridge the radiometric gap between sensors without any modification to the model weights.

Regarding the second research question, frozen encoder fine-tuning achieves the strongest performance among all inductive strategies, reaching an mF1 of

0.961 and surpassing both output-layer fine-tuning (0.952) and full fine-tuning (0.956). The best-performing strategy demonstrates that fine-tuning the decoder together with the output layer provides the optimal degree of network adaptation for cross-sensor transfer between these two sensors. However, even output layer fine-tuning alone achieves an mF1 of 0.952, which is just below the from-scratch baseline of 0.954. The result suggests that even minimal adaptation is sufficient for effective cross-sensor transfer between these two sensors. Full fine-tuning of the entire network does not provide additional benefit and slightly reduces performance compared to frozen encoder fine-tuning, though the difference is small at 0.005 mF1.

Addressing the third research question, all inductive experiments achieve performance comparable to the from-scratch baseline of mF1 0.954. Even the weakest inductive result, achieved by output layer fine-tuning without normalization, reaches an mF1 of 0.923, only 3.1 percentage points below the baseline. The best result, achieved by frozen encoder fine-tuning with fixed Z-score normalization using VEN μ S statistics, reaches an mF1 of 0.961, surpassing the from-scratch baseline by 0.007 mF1 points. This demonstrates that transfer learning from Sentinel-2 can match, and in some cases even slightly exceed, the performance of a model trained entirely on VEN μ S data, reducing the need for large amounts of manually labeled target domain data. MAJA, evaluated visually only due to its known limitation of overestimating cloud cover, clearly underperforms all deep learning approaches.

5 Discussion

The results demonstrate that successful cross-sensor transfer from Sentinel-2 to VEN μ S is possible across all investigated settings. However, the performance strongly depends on both the transfer learning strategy and the normalization approach applied. The following sections discuss the main factors influencing these results, compare the findings with related work, and identify limitations and directions for future research.

5.1 Effect of Normalization Strategy

Normalization strategy turned out to be one of the most influential variables in this study, with the influence varying across transfer learning settings. In the transductive setting, the choice of normalization strategy was the critical factor between total model failure and strong performance. Both raw and fixed Z-score with Sentinel-2 statistics lead to nearly zero cloud detection score, while dynamic Z-score achieved an mF1 of 0.919. This shows that while spectral alignment is necessary for cross-sensor transfer, it is not sufficient if absolute value ranges of the two datasets differ.

One reason for this difference lies in the difference in product level between the two datasets. The Sentinel-2 training data consists of Level 1C top-of-atmosphere reflectances, while VEN μ S Level 2 products provide atmospherically corrected surface reflectances, resulting in systematically different value ranges across sensors. Raw strategy fails because the model receives VEN μ S reflectance values that are systematically lower than those seen during Sentinel-2 training, placing inputs outside the value range the model was trained for. Fixed Z-score with Sentinel-2 statistics fails for the same reason. While it normalizes the data, it does so using statistics derived from the Sentinel-2 training distribution, meaning VEN μ S values still fall far outside the normalized range seen during training. Fixed Z-score with VEN μ S statistics partially addresses this by mapping the target data to its own distribution, but Z-score normalization only aligns the mean and standard deviation and does not account for differences in histogram shape arising from the distinct sensor characteristics and product levels of the two datasets. Dynamic Z-score effectively bypasses this problem by normalizing each tile independently to zero mean and unit standard deviation, ensuring that the model receives inputs in a consistent range regardless of the sensor or product level.

In the inductive setting, the normalization pattern shifts. Fixed Z-score with VEN μ S statistics consistently outperforms or matches dynamic Z-score across all fine-tuning strategies, though the gap between the two is small in most

cases and becomes negligible under full fine-tuning. This suggests that when sufficient target domain adaptation is applied through fine-tuning, the model is able to compensate for residual radiometric differences, making the choice between these two strategies less critical. The from-scratch results are somewhat different, with raw normalization performing best. Since all VEN μ S scenes cover a small, spectrally consistent region of Israel, the radiometric properties are similar across tiles. In this setting, per-tile normalization likely removes useful spectral information rather than harmful radiometric variation, which explains why dynamic Z-score underperforms raw normalization in the absence of cross-sensor domain shift.

Taken together, these findings suggest that the optimal normalization strategy depends on the transfer learning setting and the geographic diversity of the dataset. If a single strategy is to be selected for general use, dynamic Z-score normalization is the most robust choice, as it performs well across all experiments.

5.2 Transfer Learning Strategies

The strong performance achieved by output layer fine-tuning suggests that the feature representations learned by the encoder and decoder on Sentinel-2 data are largely transferable to the VEN μ S domain. This is consistent with the spectral similarity between the two sensors in the shared bands. The pretrained network appears to have learned cloud-related spectral and spatial patterns that remain meaningful across these sensors, requiring only a remapping of the final class boundaries to adapt to the target domain.

Frozen encoder fine-tuning outperforming full fine-tuning is consistent with the findings of Yosinski et al. [37]. Yosinski et al. suggest that lower layers learn more general and transferable features while higher layers become increasingly specialized to the source domain. Fine-tuning the decoder while keeping the encoder frozen therefore represents the optimal trade-off, where the decoder adapts to the spatial and radiometric properties of the target domain while the encoder representations are preserved. The small performance difference between frozen encoder and full fine-tuning further suggests that updating the encoder weights during fine-tuning does not provide additional benefit for this sensor pair.

An interesting observation relates to the potential role of shadow class knowledge in the pretrained model. The Sentinel-2 pretrained model was trained with explicit cloud shadow annotations, requiring the network to learn to distinguish shadows from other dark surfaces such as water bodies and shaded terrain. Even though the output layer is replaced and evaluation is performed on two classes only, this shadow-discriminative knowledge may be retained in the encoder representations and could provide an implicit advantage for cloud detection in the target domain. This hypothesis could be tested in future work by repeating all experiments with the cloud shadow class merged into the clear class in the Sentinel-2 training data, producing a 2-class source

model that lacks explicit shadow knowledge, and comparing its performance against the results of the present study. If the transfer learning models trained without shadow knowledge underperform those trained with it, this would confirm that the shadow class knowledge encoded in the pretrained model provides an advantage for cloud detection in the target domain.

An important finding of the transductive setting is that the 3-class output of the Sentinel-2 pretrained model, including cloud shadow predictions, was successfully transferred to VEN μ S imagery without any modification to the model weights. Visual inspection of prediction results using dynamic Z-score normalization showed that shadows were detected in plausible locations.

Visual inspection of the prediction examples provides additional insight into model behaviour across different land cover types. In the transductive setting, dark surfaces such as water bodies are sometimes misclassified as shadows, and bright desert surfaces are occasionally misclassified as clouds. In the inductive setting, predictions are generally accurate across all land cover types. A minor issue is observed above urban areas, where bright white rooftops are occasionally misclassified as cloud. Visual comparison also suggests that in some locations the model predictions may even be more accurate than the ground truth annotations.

A lower learning rate of 10^{-4} was evaluated for selected fine-tuning experiments as an alternative to 10^{-3} used by default. Despite being commonly recommended for fine-tuning to avoid catastrophic forgetting [20], the lower learning rate did not improve performance and resulted in significantly longer training times.

5.3 Comparison with Related Work

As introduced in Section 2.3.1, Mateo-García et al. [6] investigated cross-sensor transfer learning between Landsat 8 and Proba-V, providing the closest comparable study to this work. Their finding that models can be trained before the target satellite is launched aligns with the assumption adopted here. In the inductive setting, they found no significant improvement with fine-tuning, which contrasts with the findings of this study. Also, both sensors in their study used top-of-atmosphere reflectances as input, meaning they did not encounter the product level mismatch present in this study, which may explain why normalization strategy was not investigated in their work. Methodologically, they used spectral response functions to align and weight band contributions, while this study uses simple central wavelength matching.

Wright et al. [4], mentioned in Section 2.3.1, demonstrated that dynamic Z-score normalization enables cross-sensor generalization without any target domain adaptation. This study confirms that dynamic Z-score normalization is effective for cross-sensor transfer, identifying dynamic Z-score as the best strategy in the transductive setting and the most robust choice overall. However, while Wright et al. focused on building a single robust model for multiple

sensors, this study investigates fine-tuning strategies for targeted adaptation to a specific sensor.

5.4 Limitations and Future Work

Several limitations of this study should be acknowledged. Both the source and target models in this study are regional. The Sentinel-2 training data covers the Middle East and North Africa region, and the VEN μ S dataset covers only part of Israel. While the geographic alignment between the two datasets likely benefits transfer learning performance, it also means that the results may not generalize to other geographic regions. Nevertheless, it should be noted that the presence of desert surfaces represents a particularly challenging case for cloud detection. It would be worth investigating how the same transfer learning framework performs in a more global setting. For example using a globally trained Sentinel-2 model trained on the full CloudSEN12+ dataset as the source. However, this would be computationally expensive given the size of the full dataset. The regional scope was a deliberate choice in this study, as a globally trained source model would likely increase the domain shift in the transductive setting. Whether this increased domain shift could still be compensated by dynamic Z-score normalization alone remains an open question.

The annotation quality of the CloudSEN12+ dataset represents an additional source of uncertainty. The dataset is described as expert-labeled, yet a manual quality review was necessary to remove patches with clearly incorrect or incomplete annotations. Therefore, the annotation quality is not as high as the authors state. Visual inspection of the prediction examples suggests that in some locations the model predictions appear more accurate than the ground truth annotations, indicating that reported metrics of the Sentinel-2 pretrained model may underestimate its true performance.

This study focuses on a single source-target sensor pair, and the generalizability of the findings to other cross-sensor transfer scenarios involving different sensors or spatial resolutions remains to be investigated. A related experiment worth investigating is the performance of the same framework using only bands available on most optical sensors, such as RGB and near-infrared, which would provide insight into the minimum spectral configuration required for effective cross-sensor transfer.

Several further methodological extensions could be explored. More advanced domain adaptation techniques, including adversarial approaches, could be investigated as alternatives to input-level normalization. Joint training, which combines labeled data from both source and target domains, represents another inductive transfer learning strategy not investigated in this study. Future work could also investigate spectral adaptation approaches that remove the requirement for spectrally equivalent bands. Finally, more recent transformer-based architectures could be explored as alternatives to the U-Net used in this study, potentially offering improved performance.

6 Conclusion

This thesis investigated cross-sensor transfer learning for cloud semantic segmentation on satellite imagery, focusing on the transfer of knowledge from Sentinel-2 to VEN μ S imagery. The theoretical background of cross-sensor transfer learning for cloud detection was reviewed, source and target datasets were prepared and preprocessed, and a U-Net model was trained and evaluated under multiple normalization and transfer learning configurations. The results were compared against both a from-scratch deep learning baseline and the MAJA classical cloud detection algorithm.

Three research questions concerning the role of radiometric normalization, the required degree of network adaptation, and the competitiveness of transfer learning relative to training from scratch were addressed. These questions were investigated through experiments evaluating multiple normalization and transfer learning configurations.

The results show that spectral band alignment alone is not sufficient for cross-sensor transfer. However, dynamic Z-score normalization enables the pre-trained model to generalize well to VEN μ S data without any modification to the model weights. Regarding network adaptation, only limited adaptation is required for effective cross-sensor transfer, with output-layer fine-tuning being sufficient. Frozen encoder fine-tuning provides the optimal performance, outperforming both output-layer fine-tuning and full fine-tuning. Overall, all fine-tuning strategies achieved performance comparable to the from-scratch baseline, with frozen encoder fine-tuning surpassing it.

As part of this work, a modular preprocessing pipeline and evaluation and visualization scripts were developed and are publicly available in the thesis repository,¹ and contributions were made to the open-source cnn-lib library.² These contributions extend the functionality of cnn-lib and make the developed methods available to the broader remote sensing research community, where they can be reused and further developed by other researchers.

The main limitations of this study include the regional scope of both the source and target datasets, the focus on a single sensor pair, and the reliance on spectrally aligned input data. Future work may focus on more advanced domain adaptation techniques, more recent transformer-based architectures, and broader evaluation across different sensors and geographic regions. Overall, the results demonstrate that when the source and target domains are sufficiently similar, transfer learning can match or even exceed the performance of

¹<https://github.com/ctu-geoforall-lab-projects/dp-klimes-2026>

²<https://github.com/ctu-geoforall-lab/cnn-lib/commits/main/?author=klimesm&since=2025-12-05&until=2026-05-17>

a model trained entirely on target domain data. Consequently, this approach could be used to deploy cloud detection models to both emerging sensors and existing missions with limited training data, thereby reducing the need for large, sensor-specific annotated datasets. Furthermore, the framework could be used to generate improved cloud masks for historical scenes within the VEN μ S archive.

References

- [1] Singh, Rohit, Pal, Mahesh, and Biswas, Mantosh. “Cloud Detection Methods for Optical Satellite Imagery: A Comprehensive Review”. In: *Geomatics* 5.3 (2025). ISSN: 2673-7418. DOI: 10.3390/geomatics5030027.
- [2] Li, Zhiwei et al. “Cloud and cloud shadow detection for optical satellite imagery: Features, algorithms, validation, and prospects”. In: *ISPRS Journal of Photogrammetry and Remote Sensing* 188 (2022), pp. 89–108. ISSN: 0924-2716. DOI: 10.1016/j.isprsjprs.2022.03.020.
- [3] Zhu, Xiao Xiang et al. “Deep Learning in Remote Sensing: A Comprehensive Review and List of Resources”. In: *IEEE Geoscience and Remote Sensing Magazine* 5.4 (2017), pp. 8–36. DOI: 10.1109/MGRS.2017.2762307.
- [4] Wright, Nicholas et al. “Training sensor-agnostic deep learning models for remote sensing: Achieving state-of-the-art cloud and cloud shadow identification with OmniCloudMask”. In: *Remote Sensing of Environment* 322 (2025), p. 114694. ISSN: 0034-4257. DOI: 10.1016/j.rse.2025.114694.
- [5] Wang, Zhengxin et al. “Deep Learning-Based Cloud Detection for Optical Remote Sensing Images: A Survey”. In: *Remote Sensing* 16.23 (2024). ISSN: 2072-4292. DOI: 10.3390/rs16234583.
- [6] Mateo-García, Gonzalo et al. “Transferring deep learning models for cloud detection between Landsat-8 and Proba-V”. In: *ISPRS Journal of Photogrammetry and Remote Sensing* 160 (2020), pp. 1–17. ISSN: 0924-2716. DOI: 10.1016/j.isprsjprs.2019.11.024.
- [7] King, Michael D. et al. “Spatial and Temporal Distribution of Clouds Observed by MODIS Onboard the Terra and Aqua Satellites”. In: *IEEE Transactions on Geoscience and Remote Sensing* 51.7 (2013), pp. 3826–3852. DOI: 10.1109/TGRS.2012.2227333.
- [8] Baetens, Louis, Desjardins, Camille, and Hagolle, Olivier. “Validation of Copernicus Sentinel-2 Cloud Masks Obtained from MAJA, Sen2Cor, and FMask Processors Using Reference Cloud Masks Generated with a Supervised Active Learning Procedure”. In: *Remote Sensing* 11.4 (2019). ISSN: 2072-4292. DOI: 10.3390/rs11040433.
- [9] Hollstein, André et al. “Ready-to-Use Methods for the Detection of Clouds, Cirrus, Snow, Shadow, Water and Clear Sky Pixels in Sentinel-2 MSI Images”. In: *Remote Sensing* 8.8 (2016). ISSN: 2072-4292. DOI: 10.3390/rs8080666.
- [10] Pešek, Ondřej, Segal-Rozenhaimer, Michal, and Karnieli, Arnon. “Using Convolutional Neural Networks for Cloud Detection on VENµS Images

- over Multiple Land-Cover Types”. In: *Remote Sensing* 14.20 (2022). ISSN: 2072-4292. DOI: 10.3390/rs14205210.
- [11] Irish, Richard R. “Landsat 7 automatic cloud cover assessment”. In: *Algorithms for Multispectral, Hyperspectral, and Ultraspectral Imagery VI*. Ed. by Shen, Sylvia S. and Descour, Michael R. Vol. 4049. International Society for Optics and Photonics. SPIE, 2000, pp. 348–355. DOI: 10.1117/12.410358.
- [12] Irish, Richard R et al. “Characterization of the Landsat-7 ETM+ automated cloud-cover assessment (ACCA) algorithm”. In: *Photogrammetric engineering & remote sensing* 72.10 (2006), pp. 1179–1188.
- [13] Main-Knorn, Magdalena et al. “Sen2Cor for Sentinel-2”. In: *Image and Signal Processing for Remote Sensing XXIII*. Ed. by Bruzzone, Lorenzo. Vol. 10427. International Society for Optics and Photonics. SPIE, 2017, p. 1042704. DOI: 10.1117/12.2278218.
- [14] Ackerman, Steven A. et al. “Discriminating clear sky from clouds with MODIS”. In: *Journal of Geophysical Research: Atmospheres* 103.D24 (1998), pp. 32141–32157. DOI: 10.1029/1998JD200032.
- [15] Zhu, Zhe and Woodcock, Curtis E. “Object-based cloud and cloud shadow detection in Landsat imagery”. In: *Remote Sensing of Environment* 118 (2012), pp. 83–94. ISSN: 0034-4257. DOI: 10.1016/j.rse.2011.10.028.
- [16] Zhu, Zhe, Wang, Shixiong, and Woodcock, Curtis E. “Improvement and expansion of the Fmask algorithm: cloud, cloud shadow, and snow detection for Landsats 4–7, 8, and Sentinel 2 images”. In: *Remote Sensing of Environment* 159 (2015), pp. 269–277. ISSN: 0034-4257. DOI: 10.1016/j.rse.2014.12.014.
- [17] Hagolle, O. et al. “A multi-temporal method for cloud detection, applied to FORMOSAT-2, VEN_μS, LANDSAT and SENTINEL-2 images”. In: *Remote Sensing of Environment* 114.8 (2010), pp. 1747–1755. ISSN: 0034-4257. DOI: 10.1016/j.rse.2010.03.002.
- [18] Zhang, Zheng et al. “Cloudformer: Supplementary Aggregation Feature and Mask-Classification Network for Cloud Detection”. In: *Applied Sciences* 12.7 (2022). ISSN: 2076-3417. DOI: 10.3390/app12073221.
- [19] Luo, Dong et al. “A globally applicable deep learning model for Sentinel-2 cloud and shadow detection”. In: *Science of Remote Sensing* 12 (2025), p. 100278. ISSN: 2666-0172. DOI: 10.1016/j.srs.2025.100278.
- [20] Goodfellow, Ian, Bengio, Yoshua, and Courville, Aaron. *Deep Learning*. <http://www.deeplearningbook.org>. MIT Press, 2016.
- [21] Lecun, Y. et al. “Gradient-based learning applied to document recognition”. In: *Proceedings of the IEEE* 86.11 (1998), pp. 2278–2324. DOI: 10.1109/5.726791.
- [22] Long, Jonathan, Shelhamer, Evan, and Darrell, Trevor. “Fully convolutional networks for semantic segmentation”. In: *2015 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2015, pp. 3431–3440. DOI: 10.1109/CVPR.2015.7298965.
- [23] López-Puigdollers, Dan, Mateo-García, Gonzalo, and Gómez-Chova, Luis. “Benchmarking Deep Learning Models for Cloud Detection in

- Landsat-8 and Sentinel-2 Images”. In: *Remote Sensing* 13.5 (2021). ISSN: 2072-4292. DOI: 10.3390/rs13050992.
- [24] Mohajerani, Sorour, Krammer, Thomas A., and Saeedi, Parvaneh. *Cloud Detection Algorithm for Remote Sensing Images Using Fully Convolutional Neural Networks*. 2018. arXiv: 1810.05782 [cs.CV]. URL: <https://arxiv.org/abs/1810.05782>.
- [25] Pang, Shulin et al. “Convolutional Neural Network-Driven Improvements in Global Cloud Detection for Landsat 8 and Transfer Learning on Sentinel-2 Imagery”. In: *Remote Sensing* 15.6 (2023). ISSN: 2072-4292. DOI: 10.3390/rs15061706.
- [26] Ronneberger, Olaf, Fischer, Philipp, and Brox, Thomas. “U-Net: Convolutional Networks for Biomedical Image Segmentation”. In: *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015*. Ed. by Navab, Nassir et al. Cham: Springer International Publishing, 2015, pp. 234–241. arXiv: 1505.04597.
- [27] Badrinarayanan, Vijay, Kendall, Alex, and Cipolla, Roberto. “SegNet: A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation”. In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 39.12 (2017), pp. 2481–2495. DOI: 10.1109/TPAMI.2016.2644615. arXiv: 1511.00561.
- [28] Chen, Liang-Chieh et al. *Rethinking Atrous Convolution for Semantic Image Segmentation*. 2017. arXiv: 1706.05587.
- [29] Simonyan, Karen and Zisserman, Andrew. “Very Deep Convolutional Networks for Large-Scale Image Recognition”. In: *International Conference on Learning Representations (ICLR)*. 2015. arXiv: 1409.1556.
- [30] He, Kaiming et al. “Deep Residual Learning for Image Recognition”. In: *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016, pp. 770–778. DOI: 10.1109/CVPR.2016.90. arXiv: 1512.03385.
- [31] Hirose, Ikumi et al. “U-Net-Based Segmentation of Microscopic Images of Colorants and Simplification of Labeling in the Learning Process”. In: *Journal of Imaging* 8.7 (2022). ISSN: 2313-433X. DOI: 10.3390/jimaging8070177.
- [32] Wright, Nicholas et al. “CloudS2Mask: A novel deep learning approach for improved cloud and cloud shadow masking in Sentinel-2 imagery”. In: *Remote Sensing of Environment* 306 (2024), p. 114122. ISSN: 0034-4257. DOI: 10.1016/j.rse.2024.114122.
- [33] Milletari, Fausto, Navab, Nassir, and Ahmadi, Seyed-Ahmad. “V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation”. In: *2016 Fourth International Conference on 3D Vision (3DV)*. 2016, pp. 565–571. DOI: 10.1109/3DV.2016.79.
- [34] Al-Emadi, Sara, Yang, Yin, and Ofli, Ferda. “Analysing Satellite Imagery Classification under Spatial Domain Shift across Geographic Regions”. In: *International Journal of Computer Vision* 133 (Aug. 2025), pp. 7672–7709. DOI: 10.1007/s11263-025-02518-z.

- [35] Pan, Sinno Jialin and Yang, Qiang. “A Survey on Transfer Learning”. In: *IEEE Transactions on Knowledge and Data Engineering* 22.10 (2010), pp. 1345–1359. DOI: 10.1109/TKDE.2009.191.
- [36] Farahani, Abolfazl et al. *A Brief Review of Domain Adaptation*. 2020. arXiv: 2010.03978 [cs.LG]. URL: <https://arxiv.org/abs/2010.03978>.
- [37] Yosinski, Jason et al. “How Transferable are Features in Deep Neural Networks?” In: *Advances in Neural Information Processing Systems 27 (NeurIPS)*. MIT Press, 2014, pp. 3320–3328. arXiv: 1411.1792.
- [38] Powers, David M. W. *Evaluation: from precision, recall and F-measure to ROC, informedness, markedness and correlation*. 2020. arXiv: 2010.16061 [cs.LG]. URL: <https://arxiv.org/abs/2010.16061>.
- [39] Shtym, Tetiana et al. *KappaSet: Sentinel-2 KappaZeta Cloud and Cloud Shadow Masks*. Version 1. Sept. 2022. DOI: 10.5281/zenodo.7100327.
- [40] Francis, Alistair et al. *Sentinel-2 Cloud Mask Catalogue*. Nov. 2020. DOI: 10.5281/zenodo.4172871.
- [41] Baetens, Louis and Hagolle, Olivier. *Sentinel-2 reference cloud masks generated by an active learning method*. Oct. 2018. DOI: 10.5281/zenodo.1460961.
- [42] Luo, Dong and Zhang, Hankui. *Globally Distributed Sentinel-2 cloud and cloud shadow Dataset (GDSD)*. Version v1.0. Dec. 2024. DOI: 10.5281/zenodo.14397827.
- [43] Aybar, Cesar et al. “CloudSEN12+: The largest dataset of expert-labeled pixels for cloud and cloud shadow detection in Sentinel-2”. In: *Data in Brief* 56 (2024), p. 110852. ISSN: 2352-3409. DOI: 10.1016/j.dib.2024.110852.
- [44] Drusch, M. et al. “Sentinel-2: ESA’s Optical High-Resolution Mission for GMES Operational Services”. In: *Remote Sensing of Environment* 120 (2012). The Sentinel Missions - New Opportunities for Science, pp. 25–36. ISSN: 0034-4257. DOI: 10.1016/j.rse.2011.11.026.
- [45] Aybar, Cesar et al. “CloudSEN12, a global dataset for semantic understanding of cloud and cloud shadow in Sentinel-2”. In: *Scientific Data* 9 (Dec. 2022), p. 782. DOI: 10.1038/s41597-022-01878-2.
- [46] Salvoldi, Manuel et al. “VEN μ S-Derived NDVI and REIP at Different View Azimuth Angles”. In: *Remote Sensing* 14.1 (2022). ISSN: 2072-4292. DOI: 10.3390/rs14010184.
- [47] Theia Land Data Centre. *Venus Surface Reflectance*. Accessed: 2026-05-11. URL: <https://www.theia-land.fr/en/blog/product/venus/> (visited on 05/11/2026).
- [48] Hagolle, Olivier and Rodat, Damien. *20 years of CNES contributions to the Sentinel-2 mission*. CNES News. Accessed: 2026-05-17. July 2025. URL: <https://cnes.fr/en/news/20-years-cnes-contributions-sentinel-2-mission>.
- [49] Huang, Lei et al. “Normalization Techniques in Training DNNs: Methodology, Analysis and Application”. In: *IEEE Transactions on Pattern*

Analysis and Machine Intelligence 45.8 (2023), pp. 10173–10196. DOI:
10.1109/TPAMI.2023.3250241.

A Additional Results

This appendix provides supplementary results for all experiments described in Chapter 4. For the Sentinel-2 model training, confusion matrices for the remaining normalization strategies are provided. For the binary VEN μ S experiments, confusion matrices are omitted as they are fully determined by the recall values already reported in the metrics tables, and their visual representation adds limited value, as these strategies do not represent the best performing result. For all experiments, additional visual prediction examples are provided for the best performing normalization strategy only.

A.1 Sentinel-2 Model Training

True label	clear [1281194]	0.76	0.08	0.15
	cloud [955299]	0.15	0.67	0.18
	shadow [354317]	0.23	0.03	0.74
		clear	cloud	shadow
		Predicted label		

(A) Without normalization.

True label	clear [1281194]	0.77	0.19	0.04
	cloud [955299]	0.10	0.82	0.09
	shadow [354317]	0.16	0.08	0.76
		clear	cloud	shadow
		Predicted label		

(B) Fixed Z-score with Sentinel-2 statistics.

FIGURE A.1: Confusion matrices for the Sentinel-2 models on the Sentinel-2 test set. Values are normalized by row.

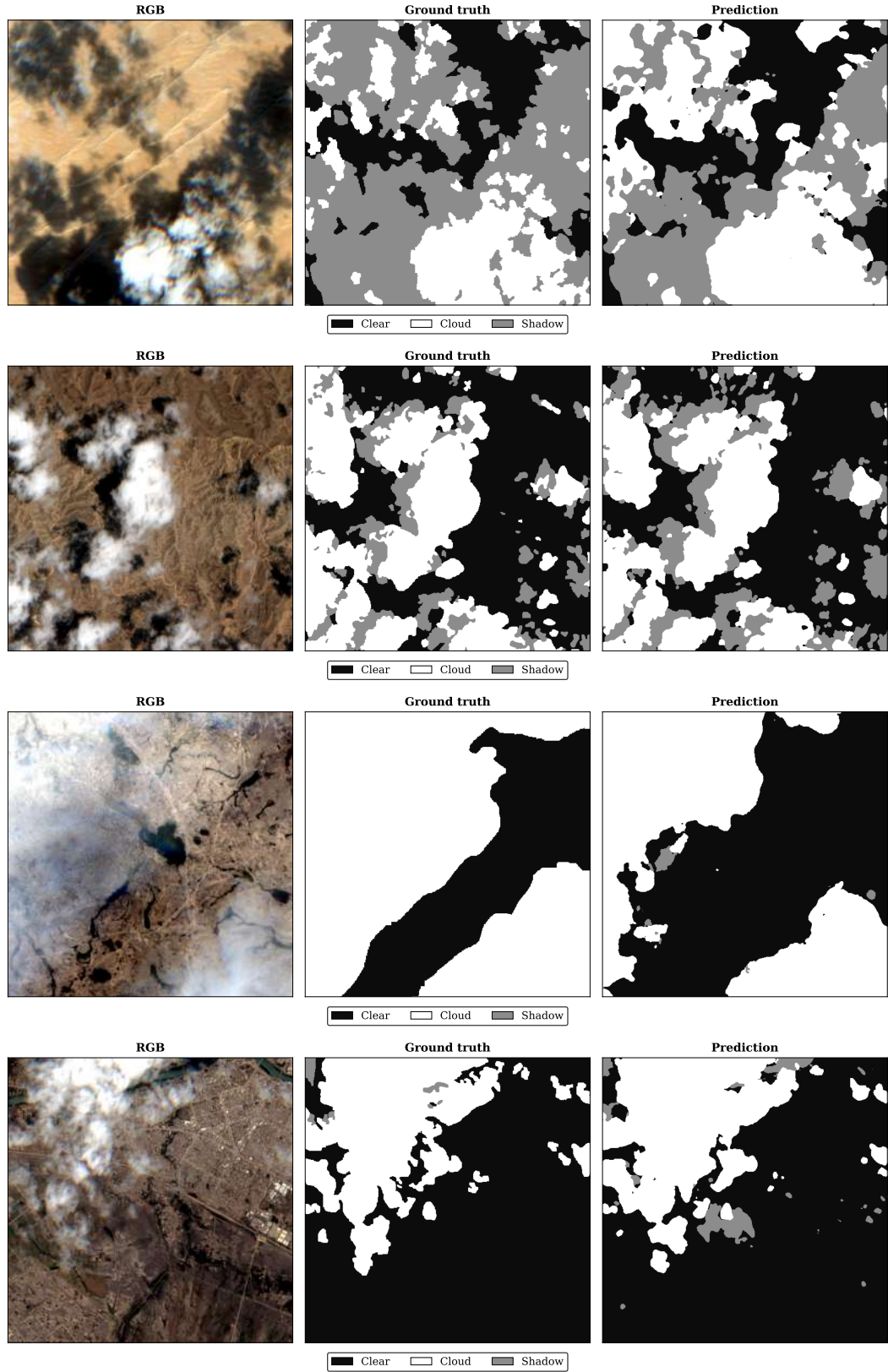


FIGURE A.2: Additional prediction examples for the Sentinel-2 model with dynamic Z-score normalization on the Sentinel-2 test set.

A.2 VEN μ S From Scratch

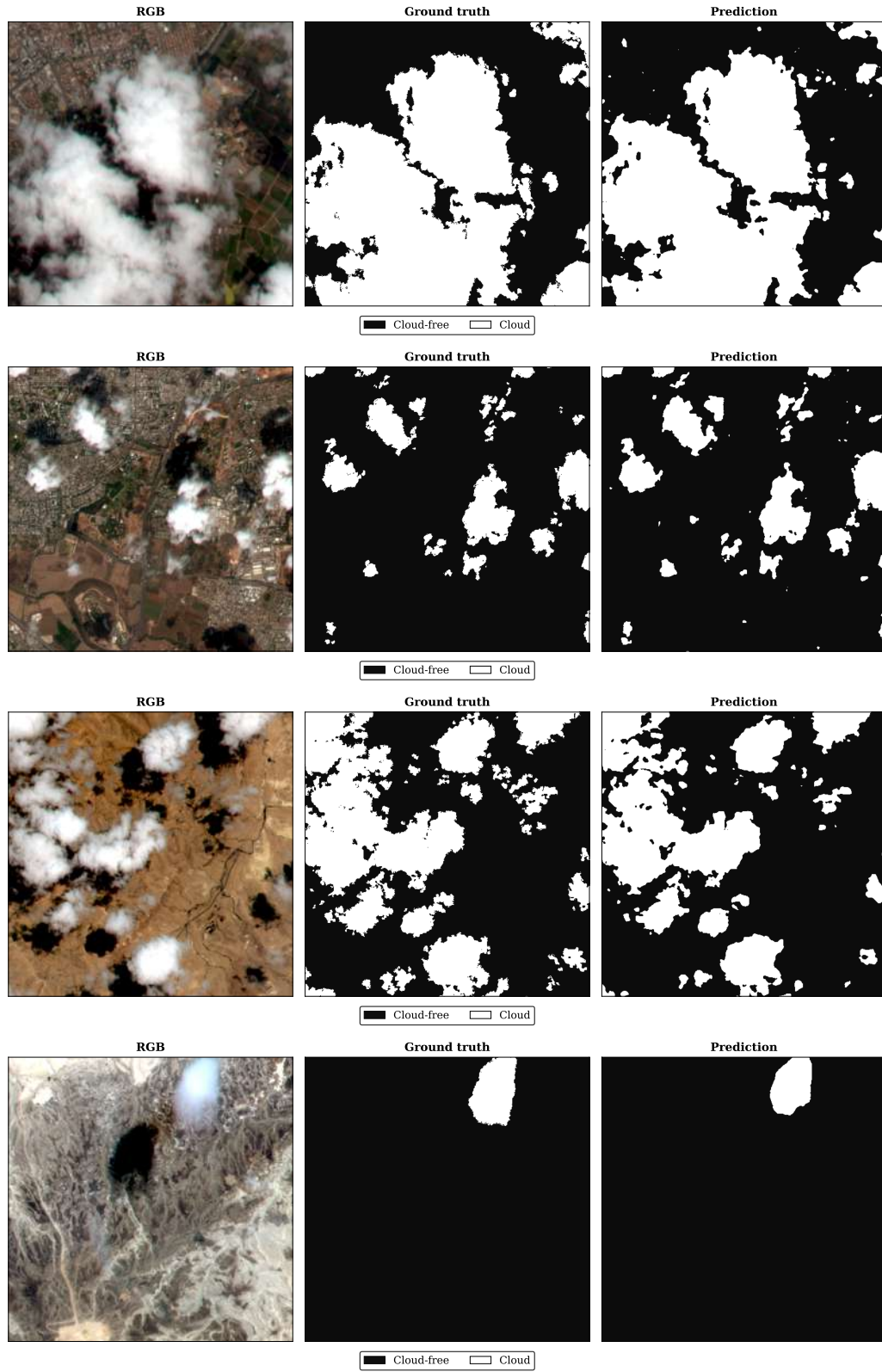


FIGURE A.3: Additional prediction examples for the VEN μ S from-scratch model without normalization on the VEN μ S test set.

A.3 Transductive Transfer Learning

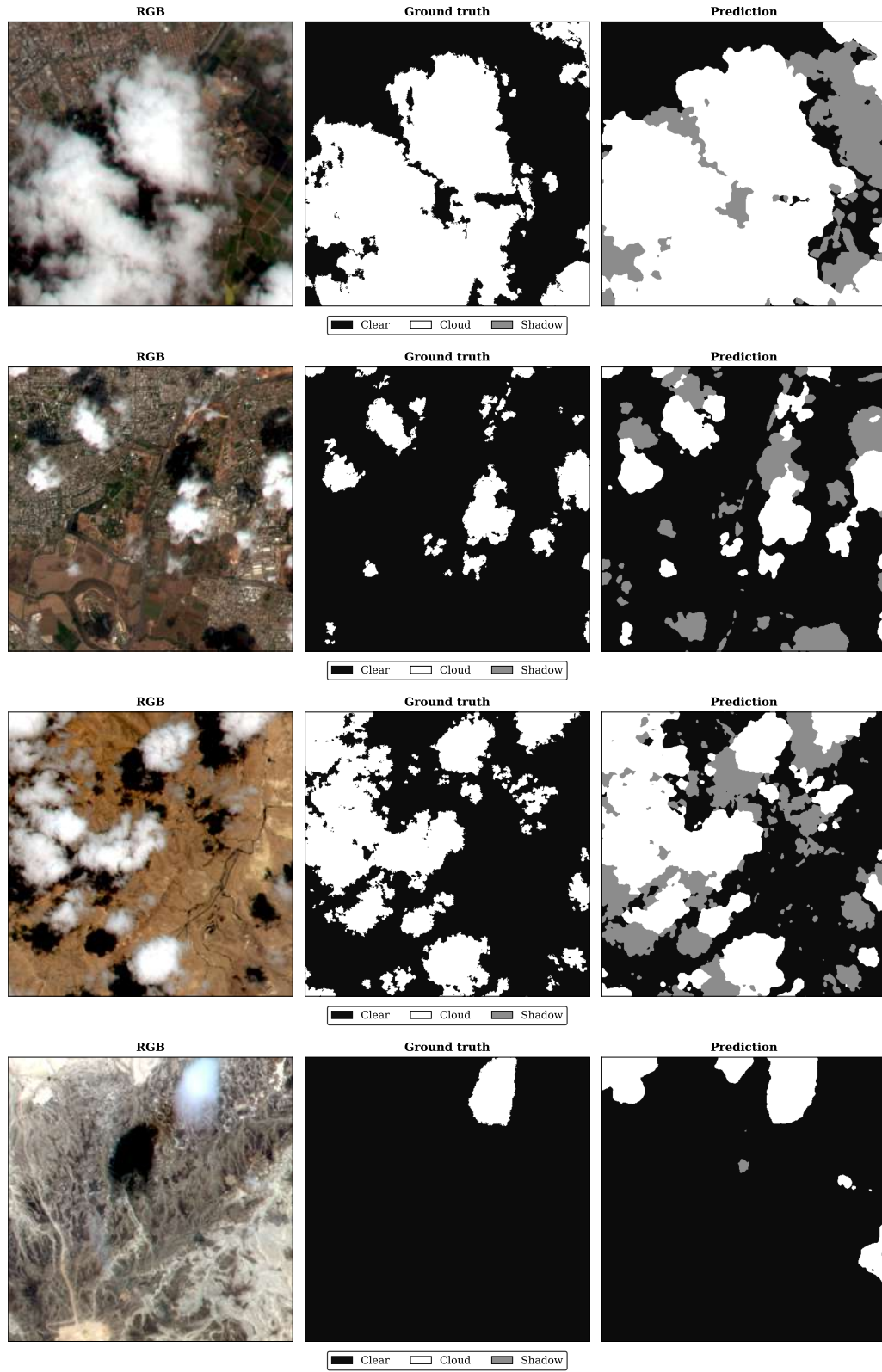


FIGURE A.4: Additional prediction examples for the transductive setting using dynamic Z-score normalization on the $\text{VEN}_{\mu\text{S}}$ test set.

A.4 Output Layer Fine-Tuning

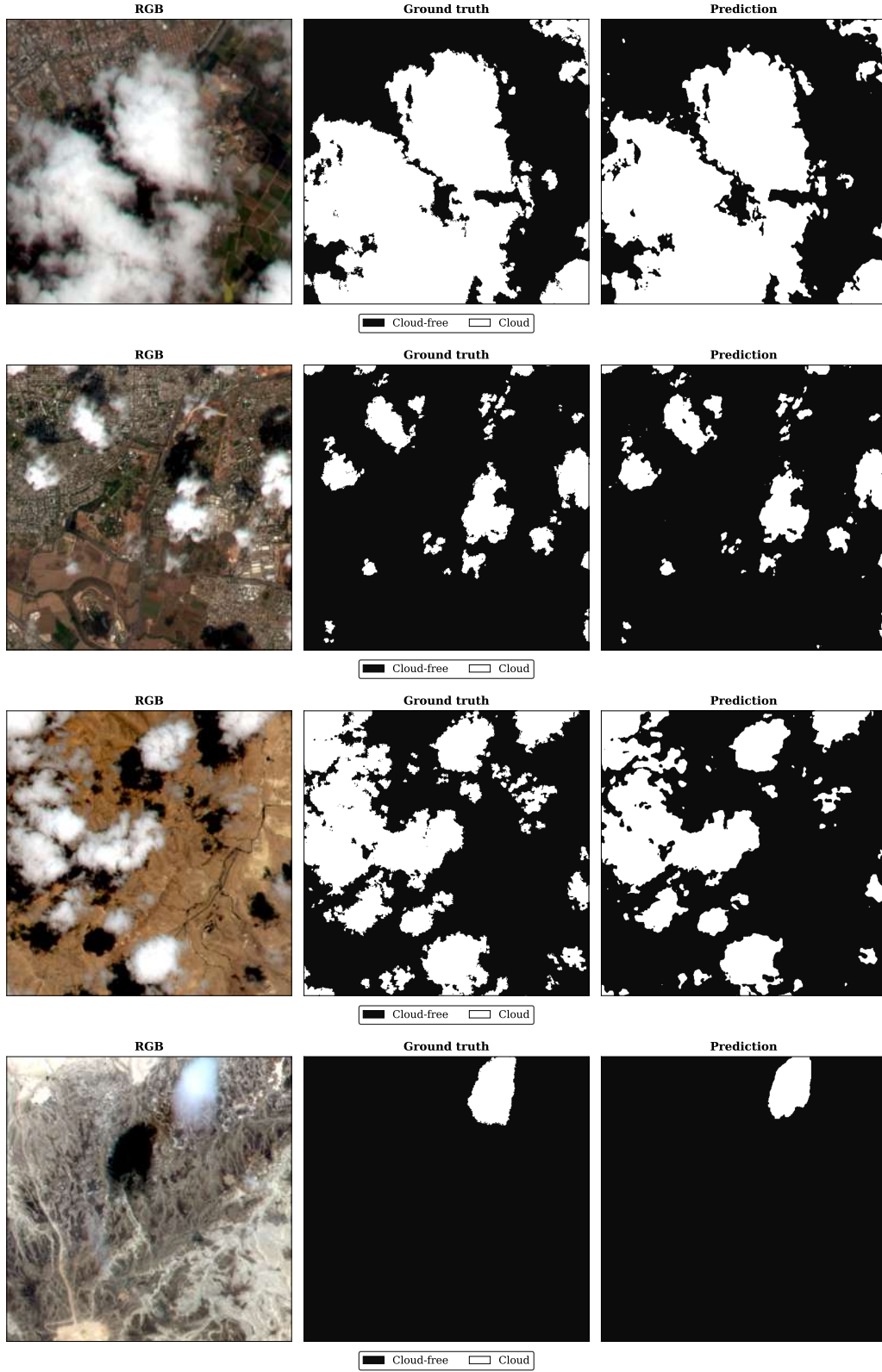


FIGURE A.5: Additional prediction examples for output layer fine-tuning using fixed Z-score normalization with $\text{VEN}_{\mu S}$ statistics on the $\text{VEN}_{\mu S}$ test set.

A.5 Frozen Encoder Fine-Tuning

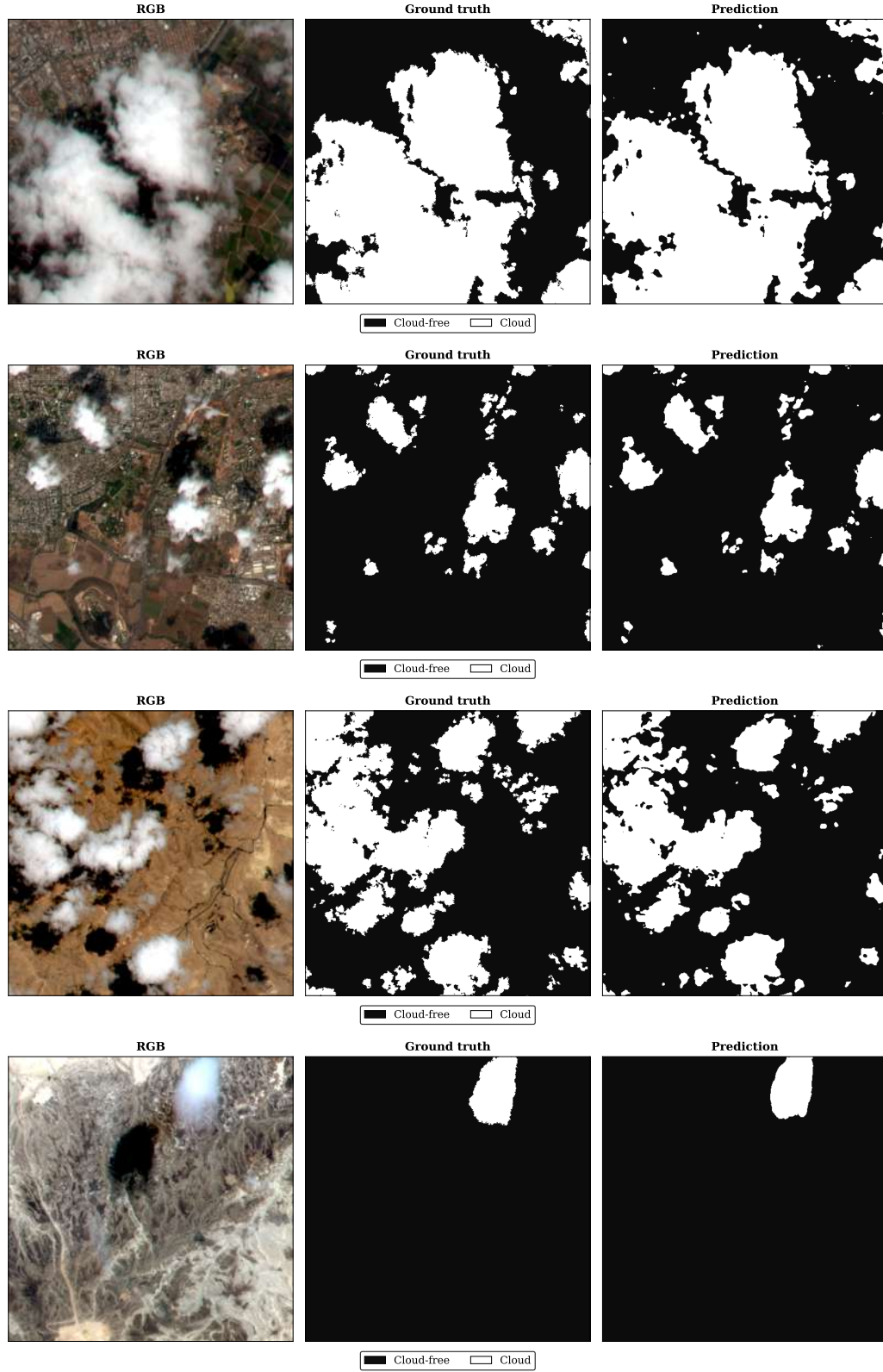


FIGURE A.6: Additional prediction examples for frozen encoder fine-tuning with fixed Z-score normalization with $\text{VEN}_{\mu S}$ statistics on the $\text{VEN}_{\mu S}$ test set.

A.6 Full Fine-Tuning

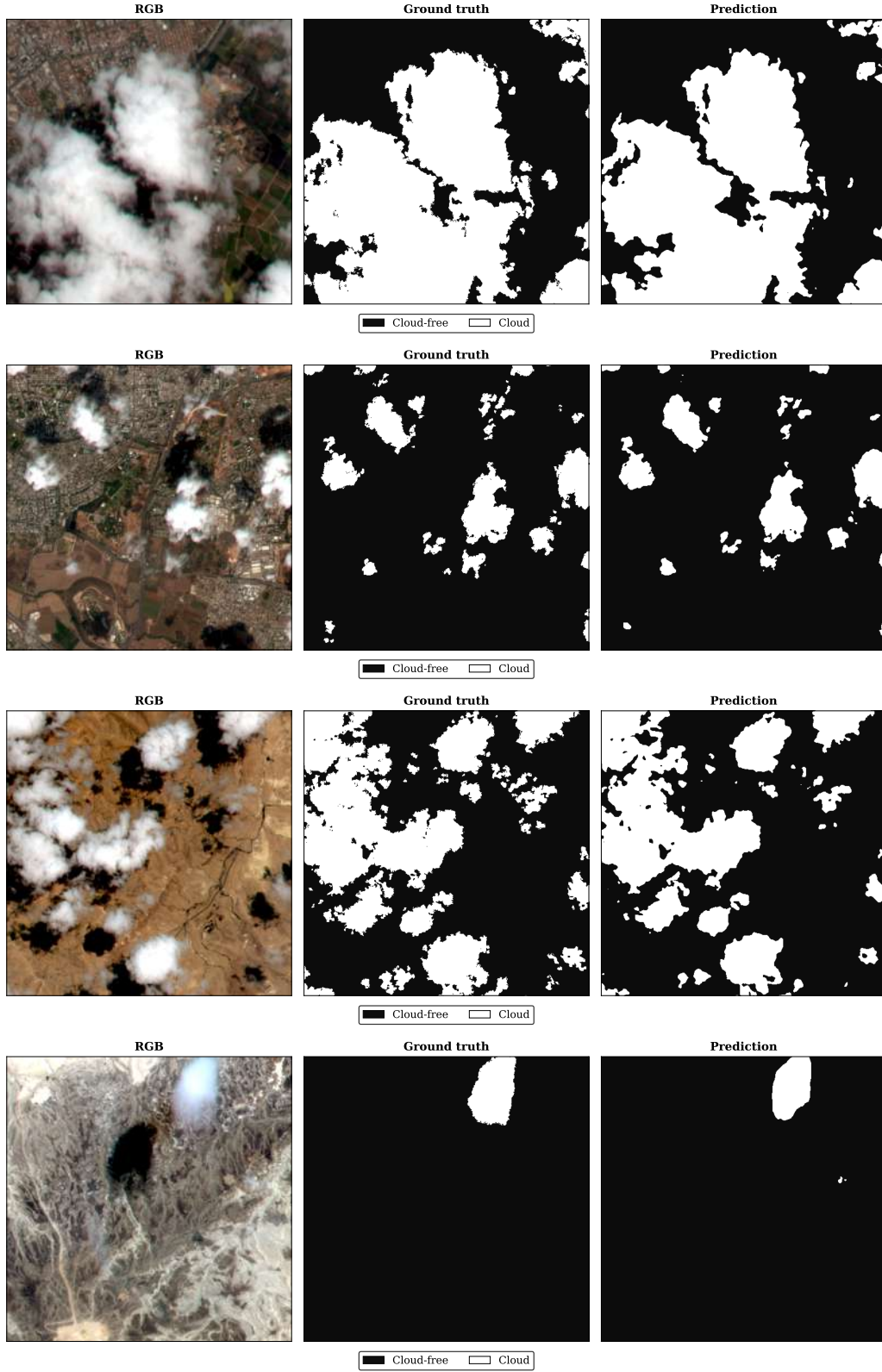


FIGURE A.7: Additional prediction examples for full fine-tuning using dynamic Z-score normalization on the VEN μ S test set.